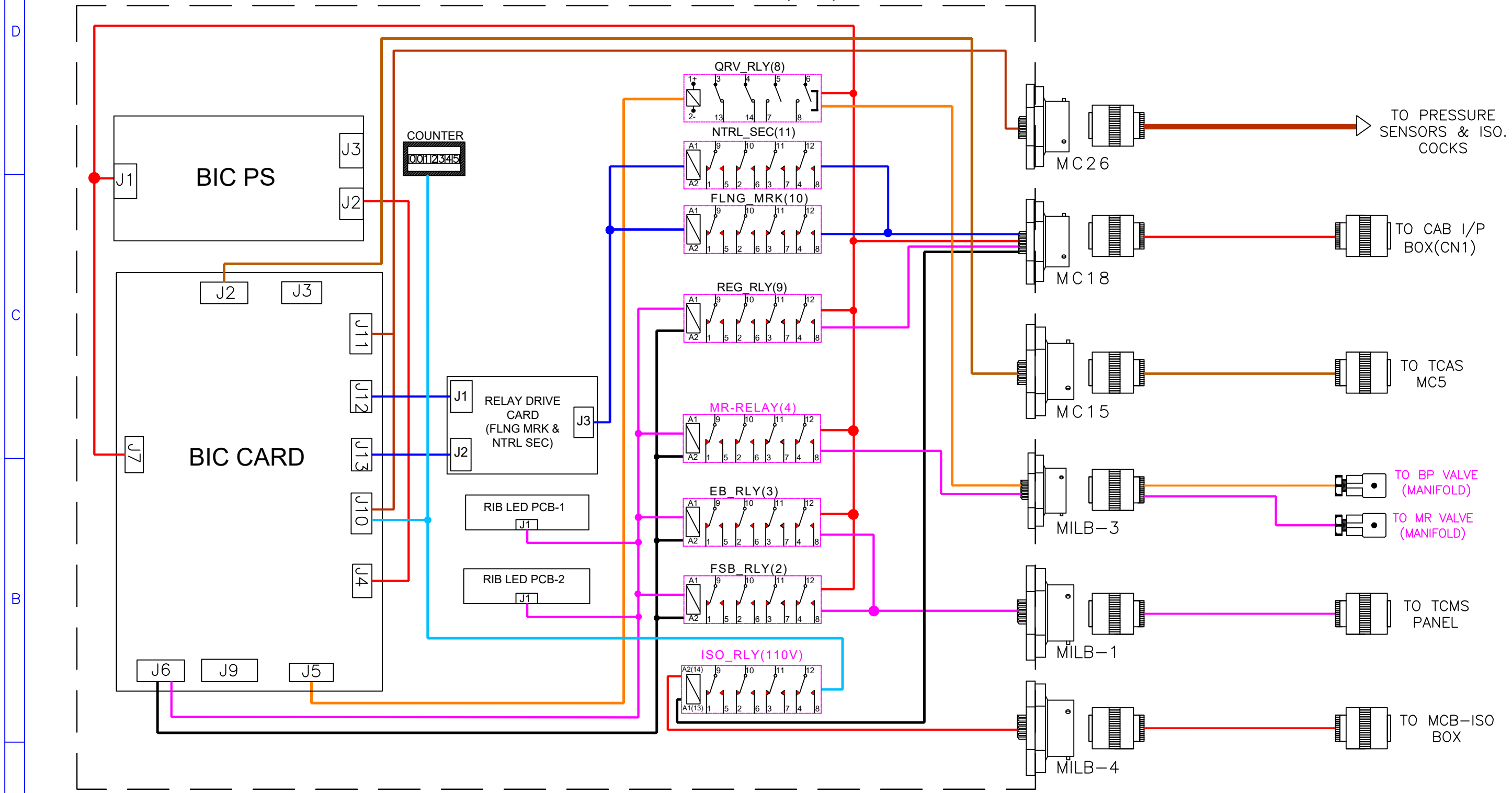


REV.NO.	CHANGE DETAILS	DATE	PREPARED	CHECKED	APPD.
1.2	DRAWING UPDATED AS PER ICF/MEDHA	24.06.26	DLR	JVP	YS
1.3	DRAWING UPDATED AS PER COE	27.07.26	DLR	JVP	YS
1.4	Relay 3&4 wiring updated in drawing	25.08.26	PGR	DLR	YS
1.5	DRAWING UPDATED as per FOAM BOARD	15.09.26	PGR	DLR	YS

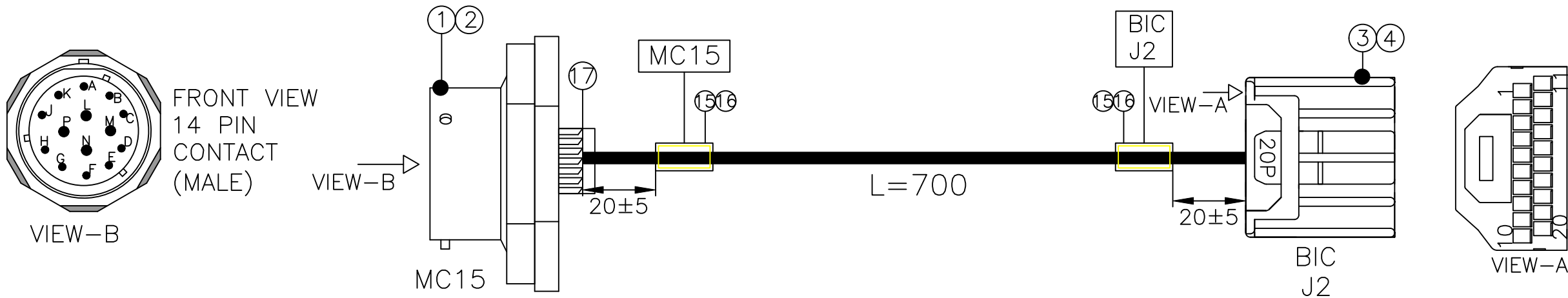
3Ph-EMU/MEMU INTERFACE UNIT (RIB)



Note:-
1.Test the all assembled cables as per testing instruction. Ref. No: 5 16 90 008

PROJECT: KAVACH		HBL HBL Engineering Ltd.	NAME		DESIGNATION	SIGN.	
ASSEMBLY: RIB-3Ph EMU			DESIGN	D.Lovaraju	Dy.Manager		
			CHECKED	J.V.Prasad	Sr.Manager		
			APPROVED	Y.Subramanyam	GM (Engg.)		
SCALE: NTS	SIZE: A3	TITLE: HARNS_INTRNAL_RIB_3PH EMU_MEMU (RIB_BLOCK DIAGRAM)			DRAWING NO. 516490494		REV. 1.5
	SHEET: 1 OF 16				PART NO. 6000054517		
EFFECTIVE FROM: ALL DIMENSIONS ARE IN MM			TOLERANCE NOT SPECIFIED SHALL BE AS PER IS : 2102 – 1993. THIS DRAWING IS THE PROPERTY OF HBL ENGINEERING LIMITED AND MUST NOT BE COPIED, REPRODUCED, TRANSMITTED OR DISCLOSED TO ANY THIRD PARTY WITHOUT THE WRITTEN CONSENT OF THE COMPANY.				

REV.NO.	CHANGE DETAILS	DATE	PREPARED	CHECKED	APPD.
1.1	UPDATED AS PER CoE/ICF REQUIREMENT	04.06.26	DLR	JVP	YS
1.2	DRAWING UPDATED AS PER ICF/MEDHA	24.06.26	DLR	JVP	YS
1.3	DRAWING UPDATED AS PER COE	27.07.26	DLR	JVP	YS
1.4	Relay_3&4 wiring updated in drawing	25.08.26	PGR	DLR	YS
1.5	DRAWING UPDATED AS PER FOAM BOARD	15.09.26	PGR	DLR	YS



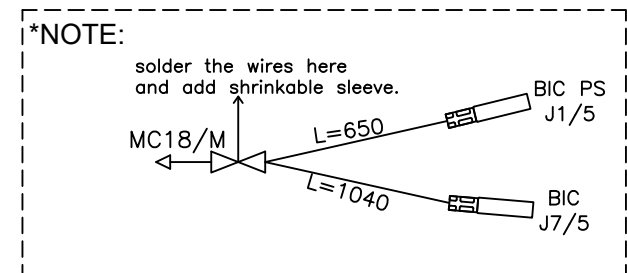
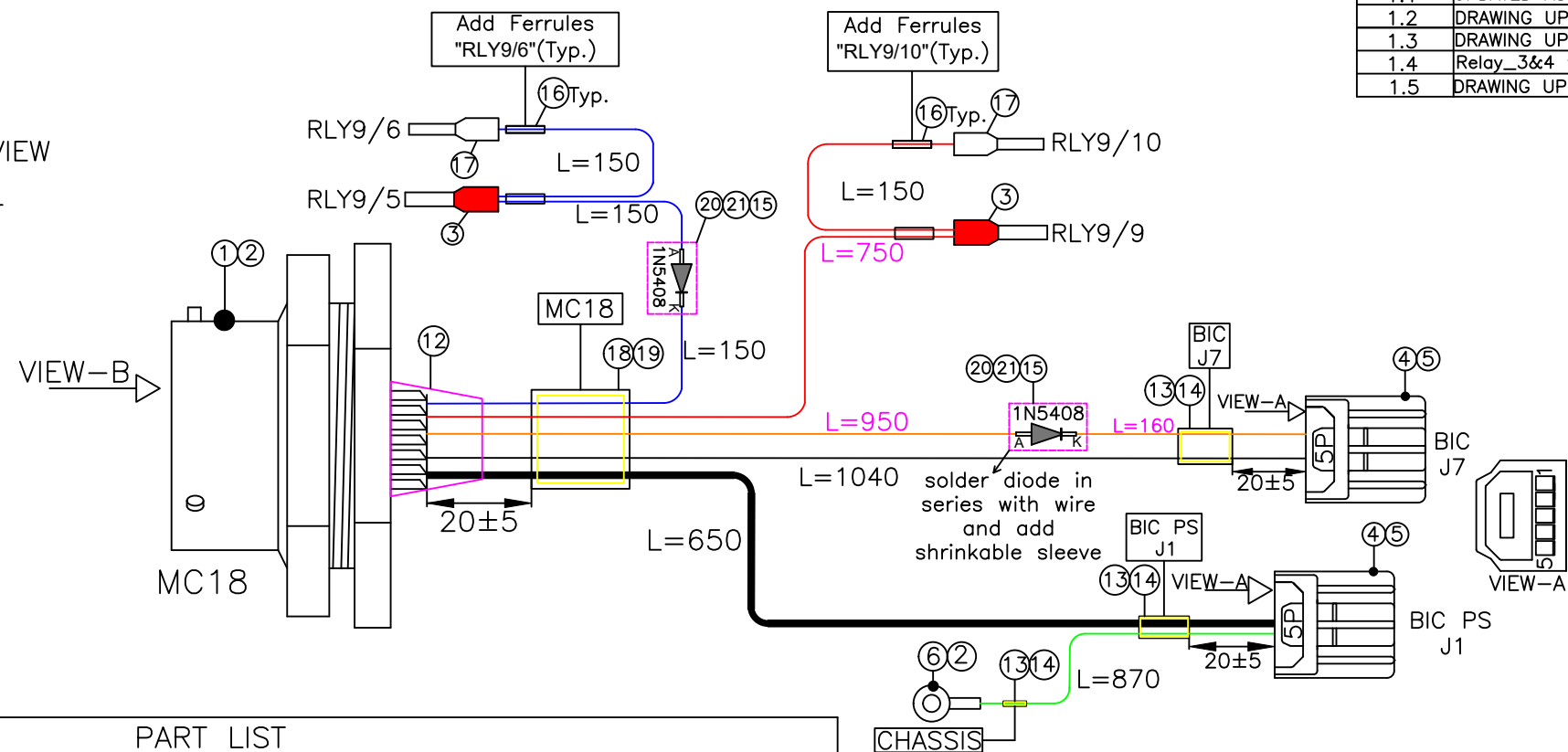
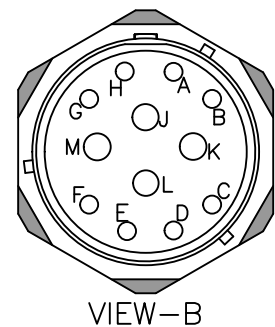
PART LIST				
ITEM NO/REF	QTY	UNITS	ITEM CODE	DESCRIPTION
1	1.00	PC	1000088342	CONNECTOR_RECEP_14P_62IN-5024-14-A14-P
2	0.20	MTR	1000016883	SLEEVE_HEAT SHINKABLE_DIA 3MM_BLACK
3	1.00	PC	1000025257	CONNECTOR_SOCKET_20POS_PNO:MX34020SF1
4	14.00	PC	1000025249	CONNECTOR_CRIMP_TERMINAL_MX34
5	1.40	MTR	1000013442	PTFE_WIRE_AWG 22(7/30)_BLACK
6	1.40	MTR	1000025380	WIRE_PTFE_BROWN_AWG 22_MIL-16878E/4BFB-1
7	1.40	MTR	1000025384	WIRE_PTFE_RED_AWG 22_MIL-16878E/4BFB-2
8	0.70	MTR	1000025383	WIRE_PTFE_ORANGE_AWG22_MIL-16878E/4BFB-3
9	1.40	MTR	1000025386	WIRE PTFE YELLOW AWG 22 MIL 16878E/4BFB-4
10	0.70	MTR	1000025381	WIRE PTFE GREEN AWG 22 MIL 16878E/4BFB-5
11	0.70	MTR	1000025379	WIRE PTFE BLUE AWG 22 MIL 16878E/4BFB-6
12	0.70	MTR	1000025385	WIRE PTFE VIOLET AWG 22 MIL 16878E/4BFB-7
13	0.70	MTR	1000025382	WIRE PTFE GREY AWG 22 MIL 16878E/4BFB-8
14	0.70	MTR	1000026453	PTFE_WIRE_AWG 22(7/30)_WHITE
15	0.10	MTR	1000016947	SLEEVE_MARKER_3/8"_YELLOW
16	0.15	MTR	1000016866	SLEEVE_CLEAR_3/8"
17	0.05	MTR	1000074458	HEAT-SHRINKABLE_SLEEVE_16MM_BLACK

WIRE TERMINATION					
MC15	BIC J2	SIGNAL	WIRE COLOUR	BOM REF#	LENGTH
L	7	DPS1_24V	RED	7	700
M	8	DPS1_GND	BLACK	5	
N	10	DPS2_24V	RED	7	
P	9	DPS2_GND	BLACK	5	
A	3	BIU_A_RX+	BROWN	6	
B	13	BIU_A_RX-	ORANGE	8	
C	12	BIU_A_TX+	GREEN	10	
D	2	BIU_A_TX-	BLUE	11	
E	1	GND_ISO1	YELLOW	9	
F	14	BIU_B_RX+	VIOLET	12	
G	4	BIU_B_RX-	GREY	13	
H	5	BIU_B_TX+	WHITE	14	
J	15	BIU_B_TX-	BROWN	6	
K	6	GND_ISO2	YELLOW	9	

Note:-
1.Label the all connectors as per respective location.

PROJECT: KAVACH		HBL HBL Engineering Ltd.		NAME	DESIGNATION	SIGN.
ASSEMBLY: RIB-3Ph EMU			DESIGN	D.Lovaraju	Dy.Manager	
			CHECKED	J.V.Prasad	Sr.Manager	
			APPROVED	Y.Subramanyam	GM (Engg.)	
SCALE: NTS	SIZE: A3	TITLE: HARNS_INTRNAL_RIB_3PH EMU_MEMU HARNS_ASY_TCAS INTERFACE SIGNALS			DRAWING NO. 516490494	REV. 1.5
	SHEET: 2 OF 16				PART NO. 6000054517	
EFFECTIVE FROM: ALL DIMENSIONS ARE IN MM			TOLERANCE NOT SPECIFIED SHALL BE AS PER IS : 2102 - 1993. THIS DRAWING IS THE PROPERTY OF HBL ENGINEERING LIMITED AND MUST NOT BE COPIED, REPRODUCED, TRANSMITTED OR DISCLOSED TO ANY THIRD PARTY WITHOUT THE WRITTEN CONSENT OF THE COMPANY.			

REV.NO.	CHANGE DETAILS	DATE	PREPARED	CHECKED	APPD.
1.1	UPDATED AS PER CoE/ICF REQUIREMENT	04.06.26	DLR	JVP	YS
1.2	DRAWING UPDATED AS PER ICF/MEDHA	24.06.26	DLR	JVP	YS
1.3	DRAWING UPDATED AS PER COE	27.07.26	DLR	JVP	YS
1.4	Relay_3&4 wiring updated in drawing	25.08.26	PGR	DLR	YS
1.5	DRAWING UPDATED AS PER FOAM BOARD	15.09.26	PGR	DLR	YS



PART LIST				
ITEM NO/REF	QTY	UNITS	ITEM CODE	DESCRIPTION
1	1.00	PC	1000088346	CONNECTOR_RECEP_12P_62IN-5024-14-12P
2	0.30	MTR	1000016883	SLEEVE_HEAT SHINKABLE_DIA 3MM_BLACK
3	2.00	PC	1000095965	FERRULE_INSULATED_1MM_58-510L8_JAINSON
4	2.00	PC	1000027695	CONNECTOR_SOCKET_5POS_MX34005SF1
5	7.00	PC	1000025249	CONNECTOR_CRIMP_TERMINAL_MX34
6	1.00	PC	1000010068	LUG_1.5SQ MM_CU_RING TYPE M6
7	2.40	MTR	1000013442	PTFE_WIRE_AWG 22(7/30)_BLACK
8	2.20	MTR	1000025384	WIRE_PTFE_RED_AWG 22_MIL-16878E/4BFB-2
9	1.65	MTR	1000025383	WIRE_PTFE_ORANGE_AWG22_MIL-16878E/4BFB-3
10	0.87	MTR	1000025381	WIRE_PTFE_GREEN_AWG 22_MIL16878E/4BFB-5
11	0.45	MTR	1000025379	WIRE_PTFE_BLUE_AWG 22_MIL-16878E/4BFB-6
12	0.05	MTR	1000074458	HEAT-SHRINKABLE_SLEEVE_16MM_BLACK
13	0.10	MTR	1000075057	HEAT_SHRINKABLE_SLEEVE_5MM_YELLOW
14	0.15	MTR	1000075058	HEAT_SHRINKABLE_SLEEVE_5MM_CLEAR
15	0.06	MTR	1000026054	SLEEVE_HEAT SHRINKABLE_5MM_BLACK
16	0.20	MTR	1000007218	FERRULE_TUBE_3.3MM_WHITE
17	2.00	PC	1000095964	FERRULE_INSULATED_0.5MM_58-508L8_JAINSON
18	0.05	MTR	1000016947	SLEEVE_MARKER_3/8"_YELLOW
19	0.06	MTR	1000016866	SLEEVE_CLEAR_3/8"
20	0.10	MTR	1000017043	SLEEVE_SHRINKABLE_3/32_BLACK
21	2.00	PC	1000075399	DIODE_SCHOTTKY_LEADED_40V/5A

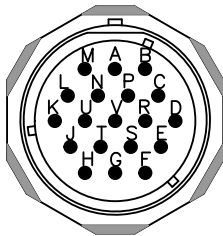
WIRE TERMINATION							
MC18	BIC PS J1	BIC J7	RLY9	SIGNAL	WIRE COLOUR	BOM REF#	LENGTH
J	1	-	-	110V_DC	RED	8	650
L	2	-	-		RED	8	
K	4	-	-	110_GND	BLACK	7	650+1040
M	5	5	-		BLACK	7	
B	-	3	-	TCR_RLY_FB	ORANGE	9	950+160
G	-	-	9&10	REG_COM	RED	8	750+150
H	-	-	5&6	REG_NO	BLUE	11	150+150+150
-	3	-	-	CHASSIS	GREEN	10	870

Note:-

1.Label the all connectors as per respective location.

PROJECT: KAVACH	HBL HBL Engineering Ltd.	NAME	DESIGNATION	SIGN.
ASSEMBLY: RIB-3Ph EMU		DESIGN	D.Lovaraju	Dy.Manager
SCALE: NTS		CHECKED	J.V.Prasad	Sr.Manager
SIZE: A3		APPROVED	Y.Subramanyam	GM (Engg.)
SHEET: 3 OF 16	TITLE: HARNS_INTRNAL_RIB_3PH EMU_MEMU HARNS ASY_BIU I/P POWER SUPPLY		DRAWING NO. 516490494	REV. 1.5
EFFECTIVE FROM:		PART NO. 6000054517		
ALL DIMENSIONS ARE IN MM		TOLERANCE NOT SPECIFIED SHALL BE AS PER IS : 2102 - 1993. THIS DRAWING IS THE PROPERTY OF HBL ENGINEERING LIMITED AND MUST NOT BE COPIED, REPRODUCED, TRANSMITTED OR DISCLOSED TO ANY THIRD PARTY WITHOUT THE WRITTEN CONSENT OF THE COMPANY.		

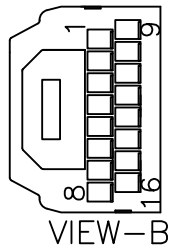
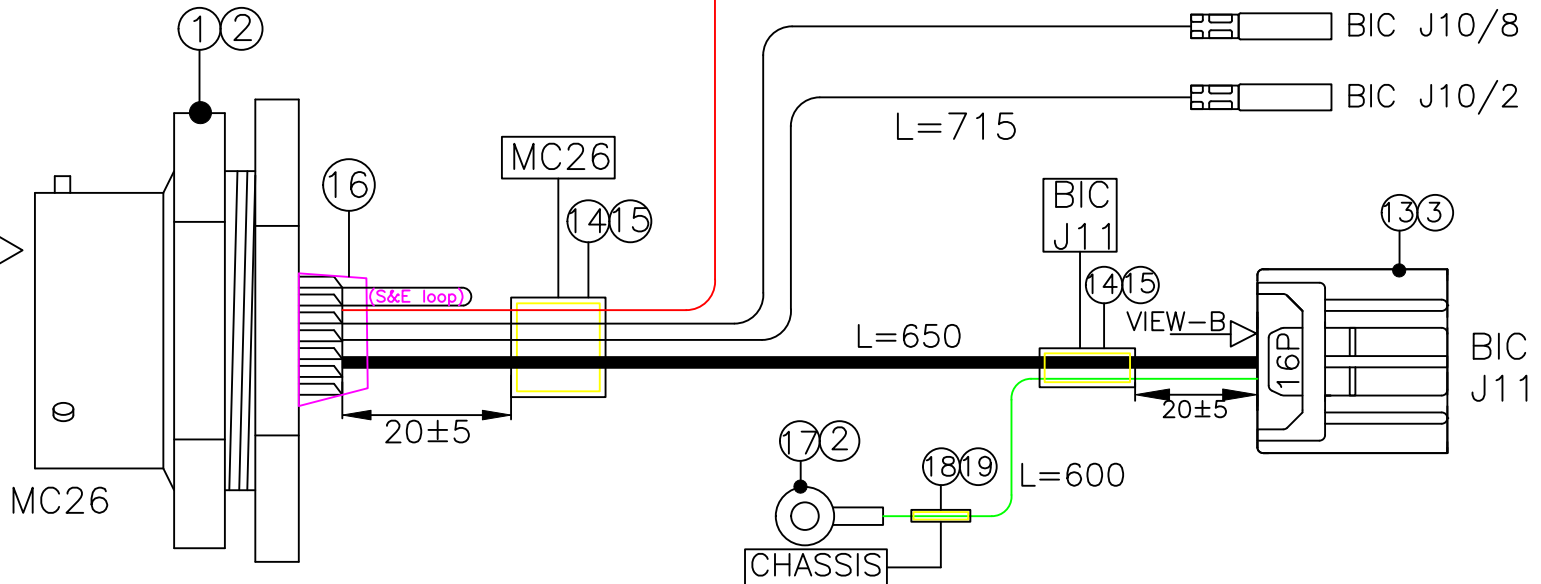
REV.NO.	CHANGE DETAILS	DATE	PREPARED	CHECKED	APPD.
1.1	UPDATED AS PER CoE/ICF REQUIREMENT	04.06.26	DLR	JVP	YS
1.2	DRAWING UPDATED AS PER ICF/MEDHA	24.06.26	DLR	JVP	YS
1.3	DRAWING UPDATED AS PER COE	27.07.26	DLR	JVP	YS
1.4	Relay_3&4 wiring updated in drawing	25.08.26	PGR	DLR	YS
1.5	DRAWING UPDATED AS PER FOAM BOARD	15.09.26	PGR	DLR	YS



FRONT VIEW
19 PIN
CONTACT
(MALE)

VIEW-A

VIEW-A



VIEW-B

PART LIST

ITEM NO/REF	QTY	UNITS	ITEM CODE	DESCRIPTION
1	1.00	PC	1000050114	CONNECTOR_PLUG_19PIN_62IN-5024-14-19P
2	0.30	MTR	1000016883	SLEEVE_HEAT SHINKABLE_DIA 3MM_BLACK
3	12.00	PC	1000025249	CONNECTOR_CRIMP_TERMINAL_MX34
4	0.72	MTR	1000013442	PTFE_WIRE_AWG 22(7/30)_BLACK
5	1.30	MTR	1000025380	WIRE_PTFE_BROWN_AWG 22_MIL-16878E/4BFB-1
6	0.65	MTR	1000025385	WIRE_PTFE_VIOLET_AWG22_MIL-16878E/4BFB-7
7	0.65	MTR	1000025383	WIRE_PTFE_ORANGE_AWG22_MIL-16878E/4BFB-3
8	0.65	MTR	1000025386	WIRE_PTFE_YELLOW_AWG 22_MIL16878E/4BFB-4
9	1.25	MTR	1000025381	WIRE_PTFE_GREEN_AWG 22_MIL16878E/4BFB-5
10	0.65	MTR	1000025379	WIRE_PTFE_BLUE_AWG 22_MIL-16878E/4BFB-6
11	0.65	MTR	1000025382	WIRE_PTFE_GREY_AWG 22_MIL-16878E/4BFB-8
12	0.65	MTR	1000026453	PTFE_WIRE_AWG 22(7/30)_WHITE
13	1.00	PC	1000025256	CONNECTOR_SOCKET_16POS_PNO:MX34016SF1
14	0.10	MTR	1000016947	SLEEVE_MARKER_3/8"_YELLOW
15	0.15	MTR	1000016866	SLEEVE_CLEAR_3/8"
16	0.05	MTR	1000074458	HEAT-SHRINKABLE_SLEEVE_16MM_BLACK
17	1.00	PC	1000010068	LUG_1.5SQ MM_CU_RING TYPE M6
18	0.05	MTR	1000075057	HEAT_SHRINKABLE_SLEEVE_5MM_YELLOW
19	0.06	MTR	1000075058	HEAT_SHRINKABLE_SLEEVE_5MM_CLEAR
20	0.72	MTR	1000025384	WIRE_PTFE_RED_AWG 22_MIL-16878E/4BFB-2

WIRE TERMINATION

MC26	BIC J11	BIC J10	SIGNAL	WIRE COLOUR	BOM REF#	LENGTH
K	4	-	MR_SENSOR_+VE	YELLOW	8	650
L	12	-	MR_PR RETURN	ORANGE	7	
A	1	-	BP_SENSOR_+VE	GREY	11	
B	9	-	BP_PR RETURN	BLUE	10	
M	5	-	BC1_SENSOR_+VE	BROWN	5	
N	13	-	BC1_PR RETURN	WHITE	12	
G	6	-	BC2_SENSOR_+VE	BROWN	5	
H	14	-	BC2_PR RETURN	VIOLET	6	
F	-	8	BP_ISO COCK	GREEN	9	715
S	-	2	24V_GND	BLACK	4	
D	-	16	MR_ISO COCK	RED	20	
E	-	2	24V_GND	BLACK	4	60(S&E loop)
-	16	-	CHASSIS	GREEN	9	600

Note:-

1.Label the all connectors as per respective location.

PROJECT: KAVACH	HBL HBL Engineering Ltd.	NAME	DESIGNATION	SIGN.
ASSEMBLY: RIB-3Ph EMU		DESIGN	D.Lovaraju	Dy.Manager
SCALE: NTS		CHECKED	J.V.Prasad	Sr.Manager
SIZE: A3		APPROVED	Y.Subramanyam	GM (Engg.)
SHEET: 4 OF 16	TITLE: HARNS_INTRNAL_RIB_3PH EMU_MEMU HARNS ASY_PRESSURE SENSORS&COCKS		DRAWING NO. 516490494	REV. 1.5
EFFECTIVE FROM:		PART NO. 6000054517		
ALL DIMENSIONS ARE IN MM		TOLERANCE NOT SPECIFIED SHALL BE AS PER IS : 2102 - 1993. THIS DRAWING IS THE PROPERTY OF HBL ENGINEERING LIMITED AND MUST NOT BE COPIED, REPRODUCED, TRANSMITTED OR DISCLOSED TO ANY THIRD PARTY WITHOUT THE WRITTEN CONSENT OF THE COMPANY.		

REV.NO.	CHANGE DETAILS	DATE	PREPARED	CHECKED	APPD.
1.1	UPDATED AS PER CoE/ICF REQUIREMENT	04.06.26	DLR	JVP	YS
1.2	DRAWING UPDATED AS PER ICF/MEDHA	24.06.26	DLR	JVP	YS
1.3	DRAWING UPDATED AS PER COE	27.07.26	DLR	JVP	YS
1.4	Relay_3&4 wiring updated in drawing	25.08.26	PGR	DLR	YS
1.5	DRAWING UPDATED AS PER FOAM BOARD	15.09.26	PGR	DLR	YS

P. Ganesan
D. L. Raju

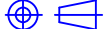
WIRE TERMINATION-1								
BIC J6	RLY4	RLY2	RLY3	RLY9	SIGNAL	WIRE COLOUR	BOM REF#	LENGTH
1*	A1	-	-	-	MR_RLY_COIL	YELLOW	9	300+220
3*	-	-	10	-	FSB_RLY_COIL	RED	7	250+220
-	-	A1	6	-	FSB_RLY_COIL	RED	7	630
5*	-	-	A1	-	EB_RLY_COIL	ORANGE	8	270+220
9	-	-	-	A1	TCR_RLY_COIL	BLUE	10	645
10	-	-	-	A2	GND_24V	BLACK	6	
2	A2	A2	A2	-	GND_24V	BLACK	6	490+270+270

WIRE TERMINATION-2						
BIC J6	D1	D2	D3	RLY4	RLY2	RLY3
1*	A	-	-	-	-	-
3*	-	A	-	-	-	-
5*	-	-	A	-	-	-
-	K	-	-	A1	-	-
-	-	K	-	-	-	10
-	-	-	K	-	-	A1

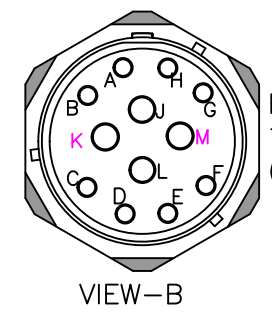
PART LIST				
ITEM NO/REF	QTY	UNITS	ITEM CODE	DESCRIPTION
1	1.00	PC	1000025257	CONNECTOR_SOCKET_20POS_PNO:MX34020SF1
2	8.00	PC	1000025249	CONNECTOR_CRIMP_TERMINAL_MX34
3	3.00	PC	1000075399	DIODE_SCHOTTKY_LEADED_40V/5A(1N5408)
4	0.10	MTR	1000026054	SLEEVE_HEAT SHRINKABLE_5MM_BLACK
5	6.00	PC	1000095965	FERRULE_INSULATED_1MM_58-510L8_JAINSON
6	1.10	MTR	1000013442	PTFE_WIRE_AWG 22(7/30)_BLACK
7	1.10	MTR	1000025384	WIRE_PTFE_RED_AWG 22_MIL-16878E/4BFB-2
8	0.49	MTR	1000025383	WIRE_PTFE_ORANGE_AWG22_MIL-16878E/4BFB-3
9	0.45	MTR	1000025386	WIRE_PTFE_YELLOW_AWG 22_MIL16878E/4BFB-4
10	0.15	MTR	1000025379	WIRE_PTFE_BLUE_AWG 22_MIL-16878E/4BFB-6
11	0.30	MTR	1000016883	SLEEVE_HEAT SHINKABLE_DIA 3MM_BLACK
12	0.05	MTR	1000075057	HEAT_SHRINKABLE_SLEEVE_5MM_YELLOW
13	0.06	MTR	1000075058	HEAT_SHRINKABLE_SLEEVE_5MM_CLEAR
14	0.20	MTR	1000007218	FERRULE_TUBE_3.3MM_WHITE
15	2.00	PC	1000037004	FERRULE_INSULATED_1.5mm_58-512L8_JAINSON
16	2.00	PC	1000095964	FERRULE_INSULATED_0.5MM_58-508L8_JAINSON

Note:-

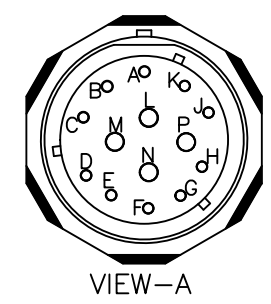
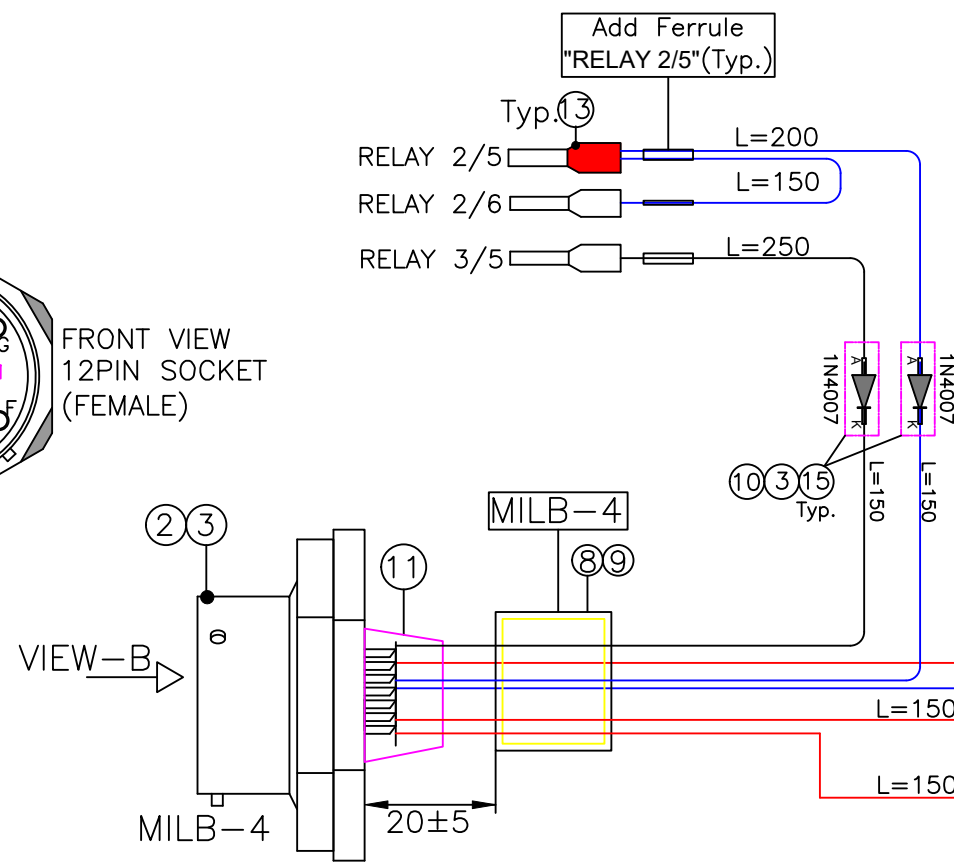
- Label the connectors as visible at respective location
- Add required ferrules to cables which are connected to relays at respective location.
- Solder the diode in between the wire and shrink with sleeve.

PROJECT: KAVACH		HBL HBL Engineering Ltd.		NAME	DESIGNATION	SIGN.	
ASSEMBLY: RIB-3Ph EMU			DESIGN	D.Lovaraju	Dy.Manager		
			CHECKED	J.V.Prasad	Sr.Manager		
			APPROVED	Y.Subramanyam	GM (Engg.)		
SCALE: NTS	SIZE: A3	TITLE: HARNS_INTRNAL_RIB_3PH EMU_MEMU HARNS ASY_RELAY DRIVE SIGNALS			DRAWING NO. 516490494		REV. 1.5
	SHEET: 5 OF 16				PART NO. 6000054517		
EFFECTIVE FROM: ALL DIMENSIONS ARE IN MM			TOLERANCE NOT SPECIFIED SHALL BE AS PER IS : 2102 - 1993.				
			THIS DRAWING IS THE PROPERTY OF HBL ENGINEERING LIMITED AND MUST NOT BE COPIED, REPRODUCED, TRANSMITTED OR DISCLOSED TO ANY THIRD PARTY WITHOUT THE WRITTEN CONSENT OF THE COMPANY.				

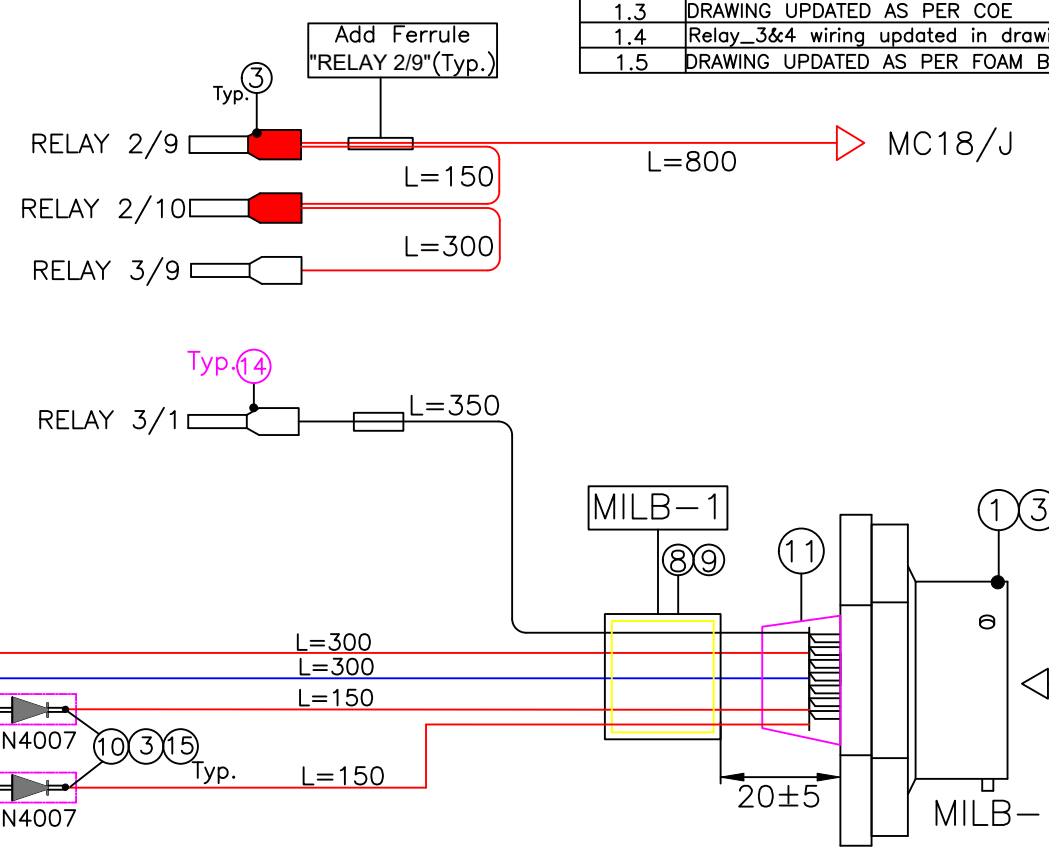
REV.NO.	CHANGE DETAILS	DATE	PREPARED	CHECKED	APPD.
1.1	UPDATED AS PER CoE/ICF REQUIREMENT	04.06.26	DLR	JVP	YS
1.2	DRAWING UPDATED AS PER ICF/MEDHA	24.06.26	DLR	JVP	YS
1.3	DRAWING UPDATED AS PER COE	27.07.26	DLR	JVP	YS
1.4	Relay_3&4 wiring updated in drawing	25.08.26	PGR	DLR	YS
1.5	DRAWING UPDATED AS PER FOAM BOARD	15.09.26	PGR	DLR	YS



FRONT VIEW
12PIN SOCKET
(FEMALE)



FRONT VIEW
14 SOCKET
CONTACT
(FEMALE)



PART LIST

ITEM NO/REF	QTY	UNITS	ITEM CODE	DESCRIPTION
1	1.00	PC	1000088343	CONNECTOR_RECEP_14P_62IN-5024-14-A14-S
2	1.00	PC	1000088347	CONNECTOR_RECEP_12P_62IN-5024-14-12S
3	0.30	MTR	1000016883	SLEEVE_HEAT SHINKABLE_DIA 3MM_BLACK
4	2.30	MTR	1000025384	WIRE_PTFE_RED_AWG 22_MIL-16878E/4BFB-2
5	0.35	MTR	1000025386	WIRE_PTFE_YELLOW_AWG 22_MIL16878E/4BFB-4
6	0.40	MTR	1000025381	WIRE_PTFE_GREEN_AWG 22_MIL16878E/4BFB-5
7	0.90	MTR	1000025379	WIRE_PTFE_BLUE_AWG 22_MIL-16878E/4BFB-6
8	0.10	MTR	1000016947	SLEEVE_MARKER_3/8"-YELLOW
9	0.12	MTR	1000016866	SLEEVE_CLEAR_3/8"
10	4.00	PC	1000006449	DIODE_LEADED_1N4007_DO-41
11	0.10	MTR	1000074458	HEAT-SHRINKABLE_SLEEVE_16MM_BLACK
12	0.50	MTR	1000007218	FERRULE_TUBE_3.3MM_WHITE
13	3.00	PC	1000095965	FERRULE_INSULATED_1MM_58-510L8_JAINSON
14	4.00	PC	1000095964	FERRULE_INSULATED_0.5MM_58-508L8_JAINSON
15	0.05	MTR	1000017038	SLEEVE_SHRINKABLE_1/16_MX446320_BLACK

- NOTE:
1. Label the all connectors as per respective location.
 2. Add ferrules to cables which are connected to relays as per respective location.

WIRE TERMINATION

MILB-1	MC18	MILB-4	RLY2	RLY3	SIGNAL	WIRE COLOUR	BOM REF#	LENGTH
B	-	A	-	-	FSB_RLY_NO	BLUE	7	300
-	-	B	5&6	-	FSB_RLY_NO	BLUE	7	200+150+150
C	-	C	-	-	EB_RLY_NO	RED	4	300
-	-	D	-	5	EB_RLY_NO	GREEN	6	250+150
D	-	-	-	1	EB_RLY_NC	YELLOW	5	350
-	J	-	9&10	9	110V_+VE	RED	4	800+150+300
L	-	G	-	-	ISO_FB_NC	RED	4	150+150
M	-	H	-	-	ISO_FB_NO	RED	4	150+150

PROJECT:
KAVACH

ASSEMBLY:
RIB-3Ph EMU

SCALE:
NTS

SIZE:
A3

SHEET:
6 OF 16

HBL
HBL Engineering Ltd.

TITLE:
HARNS_INTRNAL_RIB_3PH EMU_MEMU
HARNS_ASY_BRAKE SIGNALS

NAME
DESIGN
CHECKED
APPROVED

D.Lovaraju
J.V.Prasad
Y.Subramanyam

DESIGNATION
Dy.Manager
Sr.Manager
GM (Engg.)

DRAWING NO.
PART NO.

516490494
6000054517

REV.
1.5

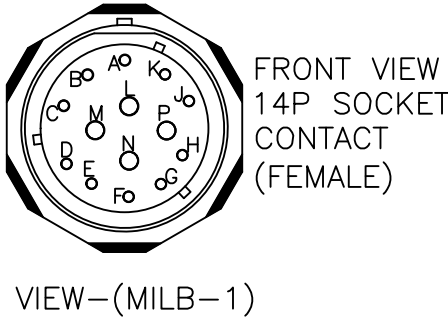
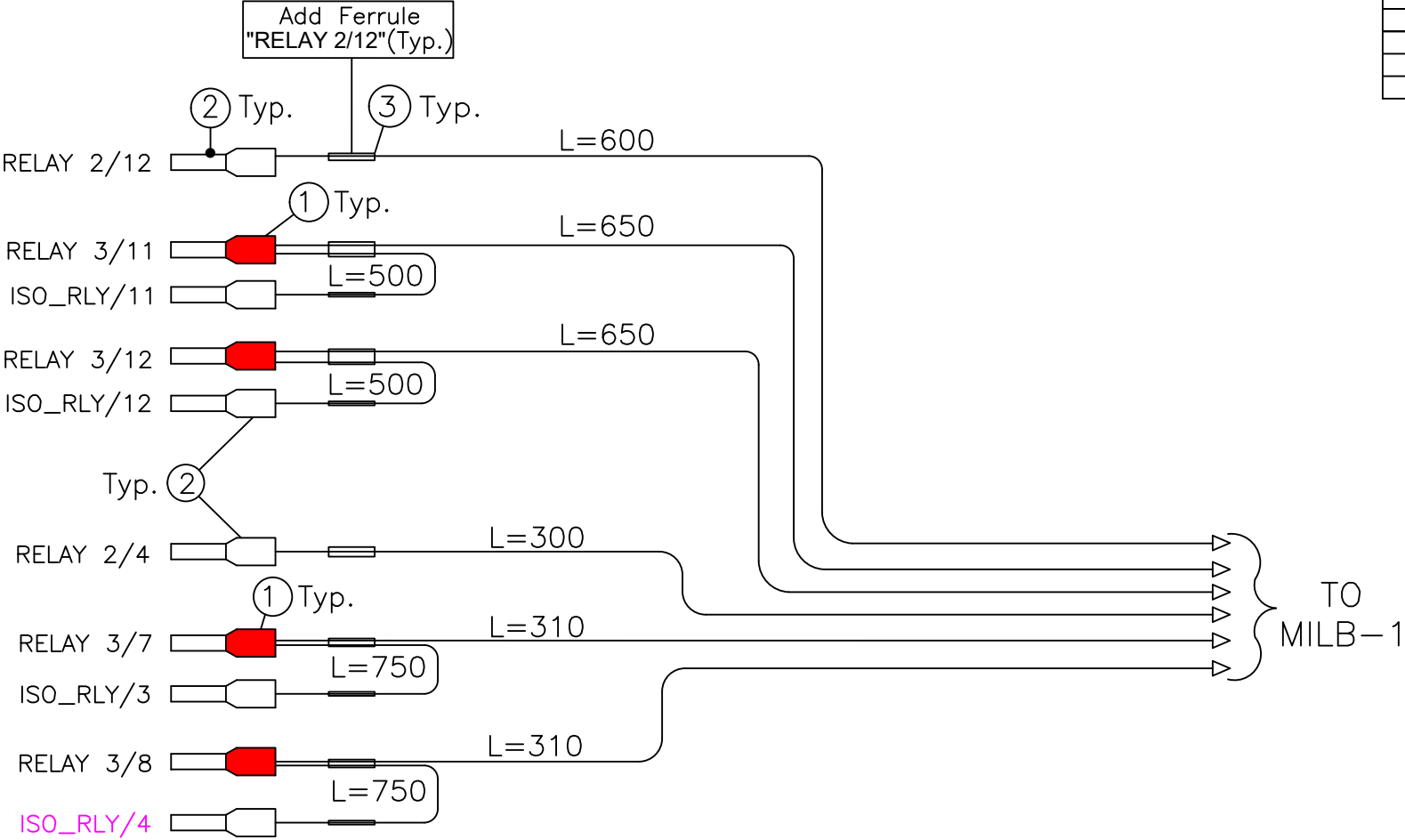
EFFECTIVE FROM:

TOLERANCE NOT SPECIFIED SHALL BE AS PER IS : 2102 - 1993.

ALL DIMENSIONS ARE IN MM

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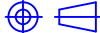
REV.NO.	CHANGE DETAILS	DATE	PREPARED	CHECKED	APPD.
1.1	UPDATED AS PER CoE/ICF REQUIREMENT	04.06.26	DLR	JVP	YS
1.2	DRAWING UPDATED AS PER ICF/MEDHA	24.06.26	DLR	JVP	YS
1.3	DRAWING UPDATED AS PER COE	27.07.26	DLR	JVP	YS
1.4	Relay_3&4 wiring updated in drawing	25.08.26	PGR	DLR	YS
1.5	DRAWING UPDATED AS PER FOAM BOARD	15.09.26	PGR	DLR	YS



PART LIST				
ITEM NO	QTY	UNITS	ITEM CODE	DESCRIPTION
1	4.00	PC	1000095965	FERRULE_INSULATED_1MM_58-510L8_JAINSON
2	6.00	PC	1000095964	FERRULE_INSULATED_0.5MM_58-508L8_JAINSON
3	0.30	MTR	1000007218	FERRULE TUBE_3.3MM_WHITE
4	0.90	MTR	1000025383	WIRE_PTFE_ORANGE_AWG22_MIL-16878E/4BFB-3
5	2.22	MTR	1000025385	WIRE_PTFE_VIOLET_AWG22_MIL-16878E/4BFB-7
6	2.22	MTR	1000025382	WIRE_PTFE_GREY_AWG 22_MIL-16878E/4BFB-8

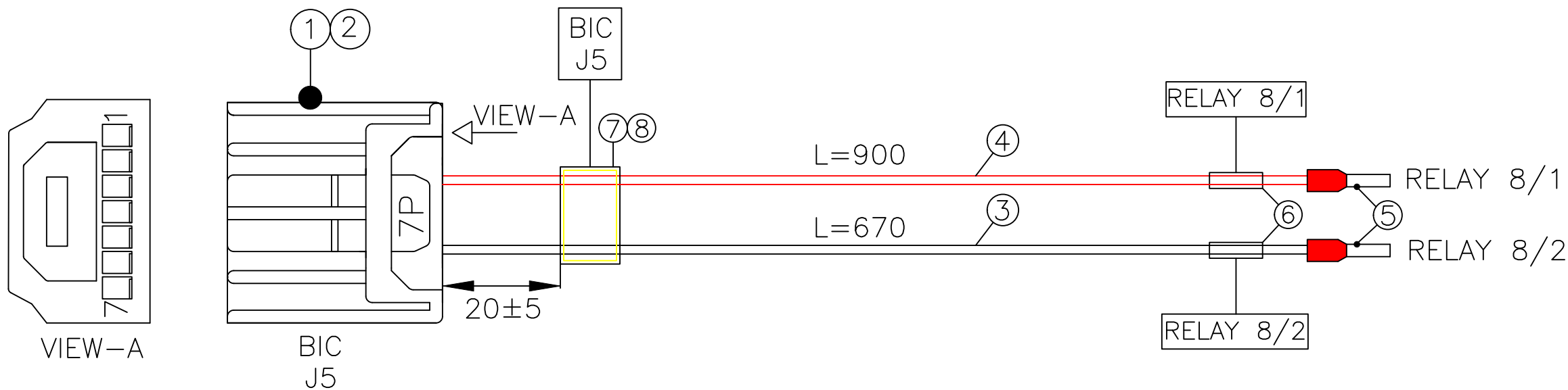
WIRE TERMINATION							
MILB-1	RLY2	RLY3	ISO RLY	SIGNAL	WIRE COLOUR	BOM REF#	LENGTH
E	12	-	-	SB_TL_NC IN	ORANGE	4	600
F	4	-	-	SB_TL_NC OUT	ORANGE	4	300
G	-	11	11	EBL_LOOP1_IN	VIOLET	5	650+500
H	-	7	3	EBL_LOOP1_OUT	VIOLET	5	310+750
J	-	12	12	EBL_LOOP2_IN	GREY	6	650+500
K	-	8	4	EBL_LOOP2_OUT	GREY	6	310+750

NOTE:
1. Label the all connectors as per respective location.
2. Add ferrules to cables which are connected to relays as per respective location.

PROJECT: KAVACH		HBL HBL Engineering Ltd.		NAME	DESIGNATION	SIGN.
ASSEMBLY: RIB-3Ph EMU			DESIGN	D.Lovaraju	Dy.Manager	
			CHECKED	J.V.Prasad	Sr.Manager	
			APPROVED	Y.Subramanyam	GM (Engg.)	
SCALE: NTS	SIZE: A3	TITLE: HARNS_INTRNAL_RIB_3PH EMU_MEMU HARNS_ASY_BRAKE SIGNALS			DRAWING NO. 516490494	REV. 1.5
	SHEET: 7 OF 16				PART NO. 6000054517	
EFFECTIVE FROM: ALL DIMENSIONS ARE IN MM			TOLERANCE NOT SPECIFIED SHALL BE AS PER IS : 2102 - 1993. THIS DRAWING IS THE PROPERTY OF HBL ENGINEERING LIMITED AND MUST NOT BE COPIED, REPRODUCED, TRANSMITTED OR DISCLOSED TO ANY THIRD PARTY WITHOUT THE WRITTEN CONSENT OF THE COMPANY.			

REV.NO.	CHANGE DETAILS	DATE	PREPARED	CHECKED	APPD.
1.1	UPDATED AS PER CoE/ICF REQUIREMENT	04.06.26	DLR	JVP	YS
1.2	DRAWING UPDATED AS PER ICF/MEDHA	24.06.26	DLR	JVP	YS
1.3	DRAWING UPDATED AS PER COE	27.07.26	DLR	JVP	YS
1.4	Relay_3&4 wiring updated in drawing	25.08.26	PGR	DLR	YS
1.5	DRAWING UPDATED AS PER FOAM BOARD	15.09.26	PGR	DLR	YS

P. Ganesan
D. L. Raju

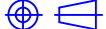


PART LIST				
ITEM NO/REF	QTY	UNITS	ITEM CODE	DESCRIPTION
1	1.00	PC	1000041208	CONNECTOR_SOCKET_7POS_MX34007SF1
2	4.00	PC	1000025249	CONNECTOR_CRIMP_TERMINAL_MX34
3	1.34	MTR	1000013442	PTFE_WIRE_AWG 22(7/30)_BLACK
4	1.80	MTR	1000025384	WIRE_PTFE_RED_AWG 22_MIL-16878E/4BFB-2
5	2.00	PC	1000037004	FERRULE_INSULATED_1.5mm_58-512L8_JAINSON
6	0.10	MTR	1000007218	FERRULE_TUBE_3.3MM_WHITE
7	0.05	MTR	1000075057	HEAT_SHRINKABLE_SLEEVE_5MM_YELLOW
8	0.06	MTR	1000075058	HEAT_SHRINKABLE_SLEEVE_5MM_CLEAR

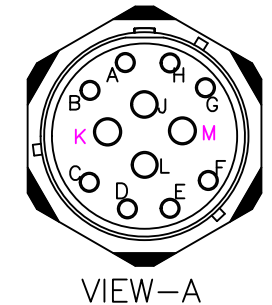
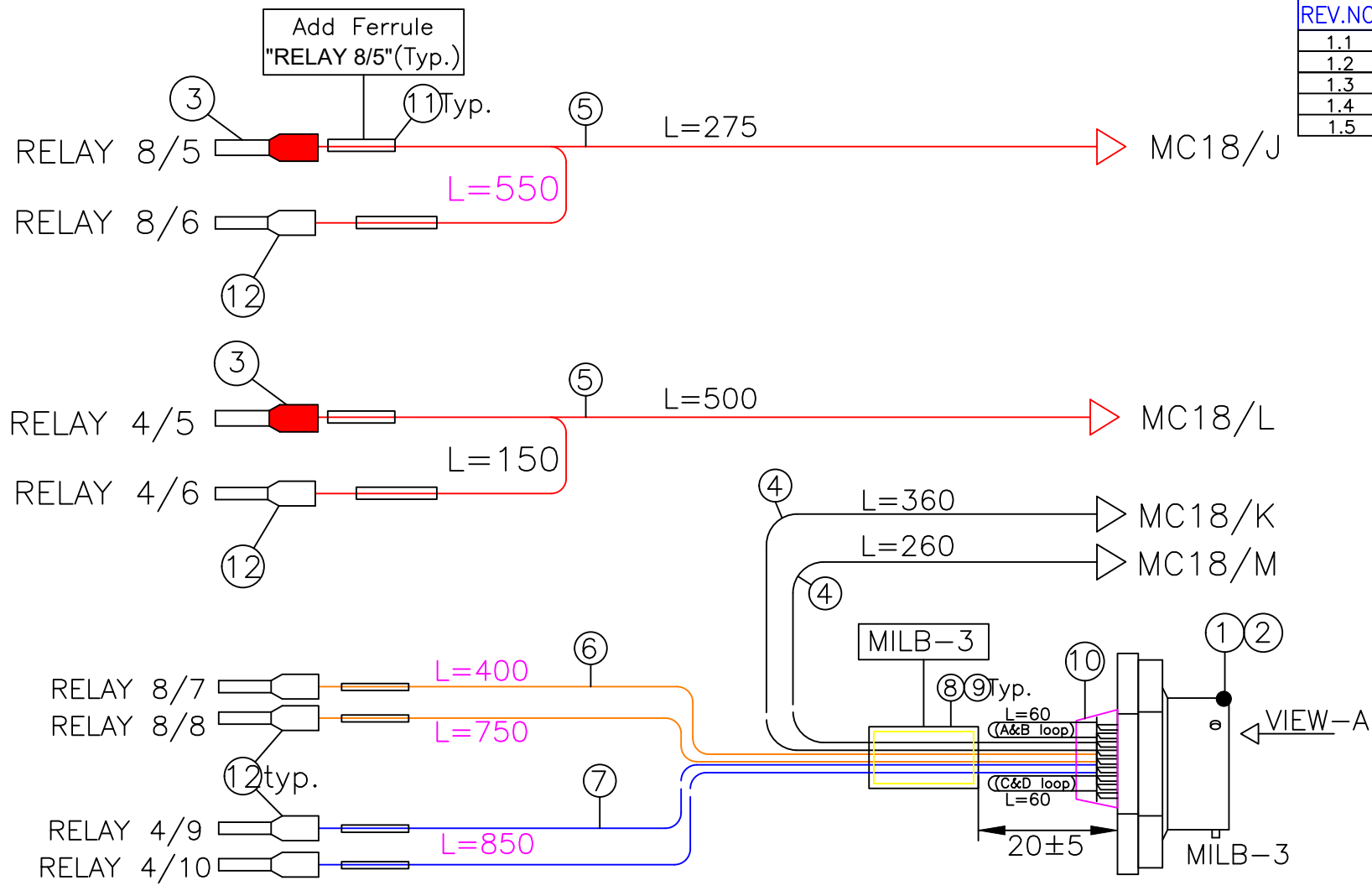
WIRE TERMINATION					
BIC J5	RELAY8	SIGNAL	WIRE COLOUR	BOM REF#	LENGTH
6,7	1	QRV_COIL_HIGH	RED	4	900
3,4	2	QRV_COIL_LOW	BLACK	3	670

Note:—

1. Label the all connectors as per respective location.
2. Add ferrules to cables which are connected to relays as per respective location.

PROJECT: KAVACH		HBL HBL Engineering Ltd.		NAME	DESIGNATION	SIGN.
ASSEMBLY: RIB-3Ph EMU			DESIGN	D.Lovaraju	Dy.Manager	
			CHECKED	J.V.Prasad	Sr.Manager	
			APPROVED	Y.Subramanyam	GM (Engg.)	
SCALE: NTS	SIZE: A3	TITLE: HARNS_INTRNAL_RIB_3PH EMU_MEMU HARNS ASY_QRV SOLENOID VALVE			DRAWING NO. 516490494	REV. 1.5
	SHEET: 8 OF 16				PART NO. 6000054517	
EFFECTIVE FROM: ALL DIMENSIONS ARE IN MM			TOLERANCE NOT SPECIFIED SHALL BE AS PER IS : 2102 - 1993. THIS DRAWING IS THE PROPERTY OF HBL ENGINEERING LIMITED AND MUST NOT BE COPIED, REPRODUCED, TRANSMITTED OR DISCLOSED TO ANY THIRD PARTY WITHOUT THE WRITTEN CONSENT OF THE COMPANY.			

REV.NO.	CHANGE DETAILS	DATE	PREPARED	CHECKED	APPD.
1.1	UPDATED AS PER CoE/ICF REQUIREMENT	04.06.26	DLR	JVP	YS
1.2	DRAWING UPDATED AS PER ICF/MEDHA	24.06.26	DLR	JVP	YS
1.3	DRAWING UPDATED AS PER COE	27.07.26	DLR	JVP	YS
1.4	Relay_3&4 wiring updated in drawing	25.08.26	PGR	DLR	YS
1.5	DRAWING UPDATED AS PER FOAM BOARD	15.09.26	PGR	DLR	YS



FRONT VIEW
12PIN SOCKET
(FEMALE)

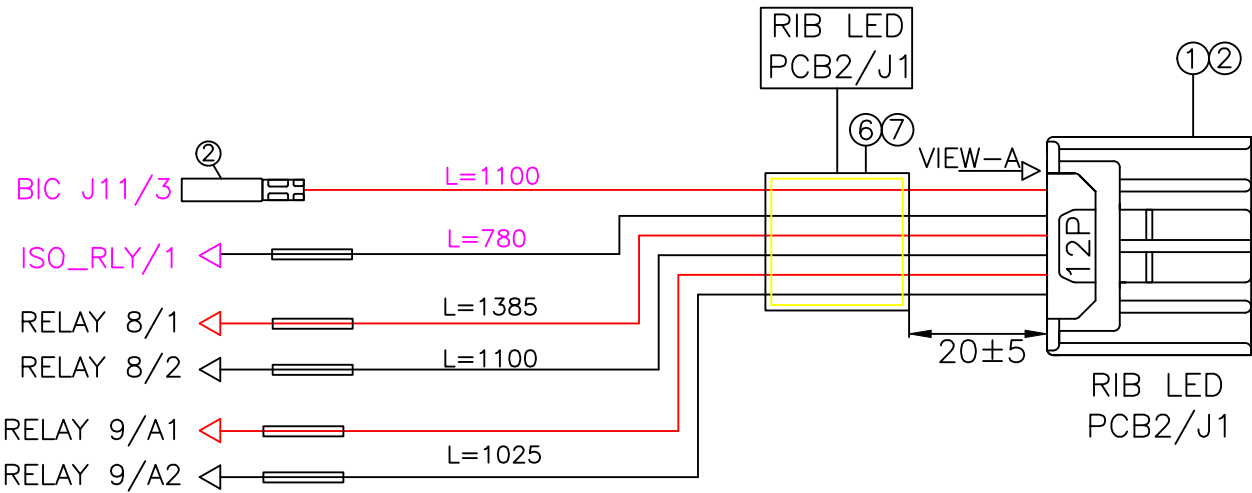
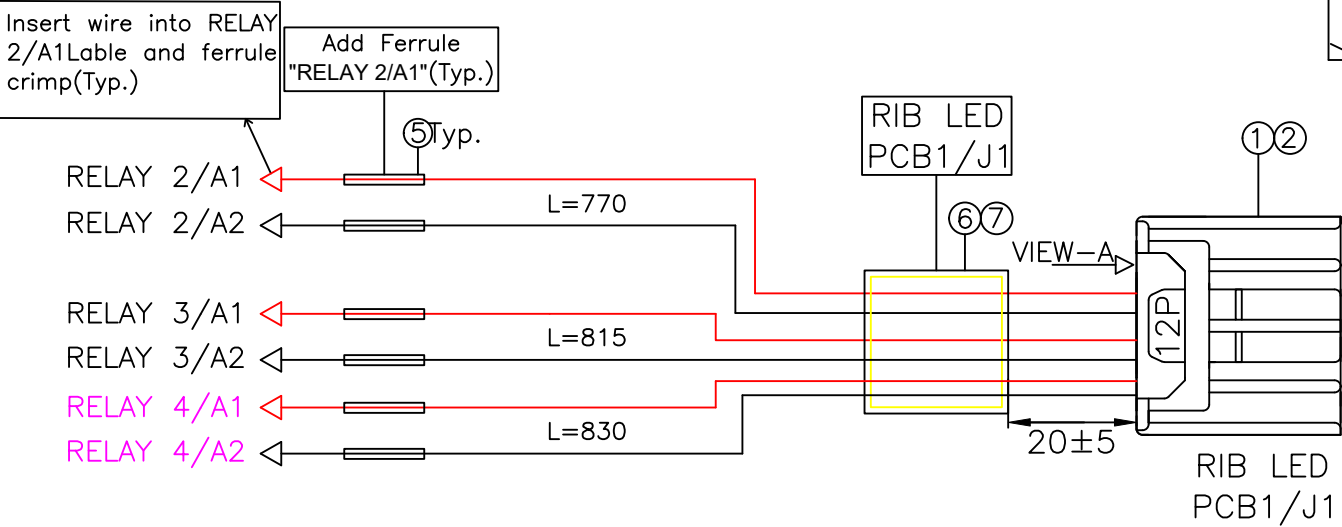
PART LIST				
ITEM NO	QTY	UNITS	ITEM CODE	DESCRIPTION
1	1.00	PC	1000088347	CONNECTOR_RECEP_12P_62IN-5024-14-12S
2	0.30	MTR	1000016883	SLEEVE_HEAT SHINKABLE_DIA 3MM_BLACK
3	2.00	PC	1000095965	FERRULE_INSULATED_1MM_58-510L8_JAINSON
4	0.53	MTR	1000013442	PTFE_WIRE_AWG 22(7/30)_BLACK
5	1.54	MTR	1000025384	WIRE_PTFE_RED_AWG 22_MIL-16878E/4BFB-2
6	1.15	MTR	1000025383	WIRE_PTFE_ORANGE_AWG22_MIL-16878E/4BFB-3
7	1.70	MTR	1000025379	WIRE_PTFE_BLUE_AWG 22_MIL-16878E/4BFB-6
8	0.05	MTR	1000016947	SLEEVE_MARKER_3/8"_YELLOW
9	0.06	MTR	1000016866	SLEEVE_CLEAR_3/8"
10	0.05	MTR	1000074458	HEAT-SHRINKABLE_SLEEVE_16MM_BLACK
11	0.20	MTR	1000007218	FERRULE_TUBE_3.3MM_WHITE
12	6.00	PC	1000095964	FERRULE_INSULATED_0.5MM_58-508L8_JAINSON

WIRE TERMINATION							
MILB-3	MC18	RELAY8	RELAY4	SIGNAL	WIRE COLOUR	BOM REF#	LENGTH
J	-	7	-	BP_VALVE_+VE	ORANGE	6	400
K	-	8	-	BP_VALVE_+VE	ORANGE	6	750
L	-	-	9	MR_VALVE_+VE	BLUE	7	850
M	-	-	10	MR_VALVE_+VE	BLUE	7	
A&B	K	-	-	BP_VALVE_-VE	BLACK	4	360+60
C&D	M	-	-	MR_VALVE_-VE	BLACK	4	260+60
-	J	5&6	-	LOCO_+VE	RED	5	275+550
-	L	-	5&6	LOCO_+VE	RED	5	500+150

Note:
1. Label the all connectors as per respective location.
2. Add ferrules to cables which are connected to relays as per respective location.

PROJECT: KAVACH	HBL HBL Engineering Ltd.	NAME	DESIGNATION	SIGN.
ASSEMBLY: RIB-3Ph EMU		DESIGN	D.Lovaraju	Dy.Manager
		CHECKED	J.V.Prasad	Sr.Manager
SCALE: NTS	SIZE: A3	APPROVED	Y.Subramanyam	GM (Engg.)
SHEET: 9 OF 16		TITLE: HARNS_INTRNAL_RIB_3PH EMU_MEMU HARNS ASY_QRV SOLENOID VALVE		DRAWING NO. 516490494 PART NO. 6000054517
EFFECTIVE FROM:		TOLERANCE NOT SPECIFIED SHALL BE AS PER IS : 2102 - 1993.		
ALL DIMENSIONS ARE IN MM		THIS DRAWING IS THE PROPERTY OF HBL ENGINEERING LIMITED AND MUST NOT BE COPIED, REPRODUCED, TRANSMITTED OR DISCLOSED TO ANY THIRD PARTY WITHOUT THE WRITTEN CONSENT OF THE COMPANY.		

REV.NO.	CHANGE DETAILS	DATE	PREPARED	CHECKED	APPD.
1.1	UPDATED AS PER CoE/ICF REQUIREMENT	04.06.26	DLR	JVP	YS
1.2	DRAWING UPDATED AS PER ICF/MEDHA	24.06.26	DLR	JVP	YS
1.3	DRAWING UPDATED AS PER COE	27.07.26	DLR	JVP	YS
1.4	Relay_3&4 wiring updated in drawing	25.08.26	PGR	DLR	YS
1.5	DRAWING UPDATED AS PER FOAM BOARD	15.09.26	PGR	DLR	YS



PART LIST				
ITEM NO/REF	QTY	UNITS	ITEM CODE	DESCRIPTION
1	2.00	PC	1000027692	CONNECTOR_SOCKET_12POS_MX34012SF1
2	21.00	PC	1000025249	CONNECTOR_CRIMP_TERMINAL_MX34
3	5.10	MTR	1000025384	WIRE_PTFE_RED_AWG 22_MIL-16878E/4BFB-2
4	4.00	MTR	1000013442	PTFE_WIRE_AWG 22(7/30)_BLACK
5	0.30	MTR	1000007218	FERRULE_TUBE_3.3MM_WHITE
6	0.10	MTR	1000016947	SLEEVE_MARKER_3/8"_YELLOW
7	0.15	MTR	1000016866	SLEEVE_CLEAR_3/8"

WIRE TERMINATION						
RIB LED PCB1/J1	RIB LED PCB2/J1	TO Connect.	SIGNAL	WIRE COLOUR	BOM REF#	LENGTH
2	—	RLY2/A1	CAB-A_FSB	RED	3	770
8	—	RLY2/A2		BLACK	4	
3	—	RLY3/A1	CAB-A_EB	RED	3	815
9	—	RLY3/A2		BLACK	4	
4	—	RLY4/A1	MR_VALVE (CAB-B_NB)	RED	3	830
10	—	RLY4/A2		BLACK	4	
—	3	BIC J11/3	ISO_MODE	RED	3	1100
—	9	ISO_RLY/1		BLACK	4	780
—	4	RLY8/1	QRV_PICKUP	RED	3	1385
—	10	RLY8/2		BLACK	4	1100
—	5	RLY9/A1	T.CUT-OFF	RED	3	1025
—	11	RLY9/A2		BLACK	4	

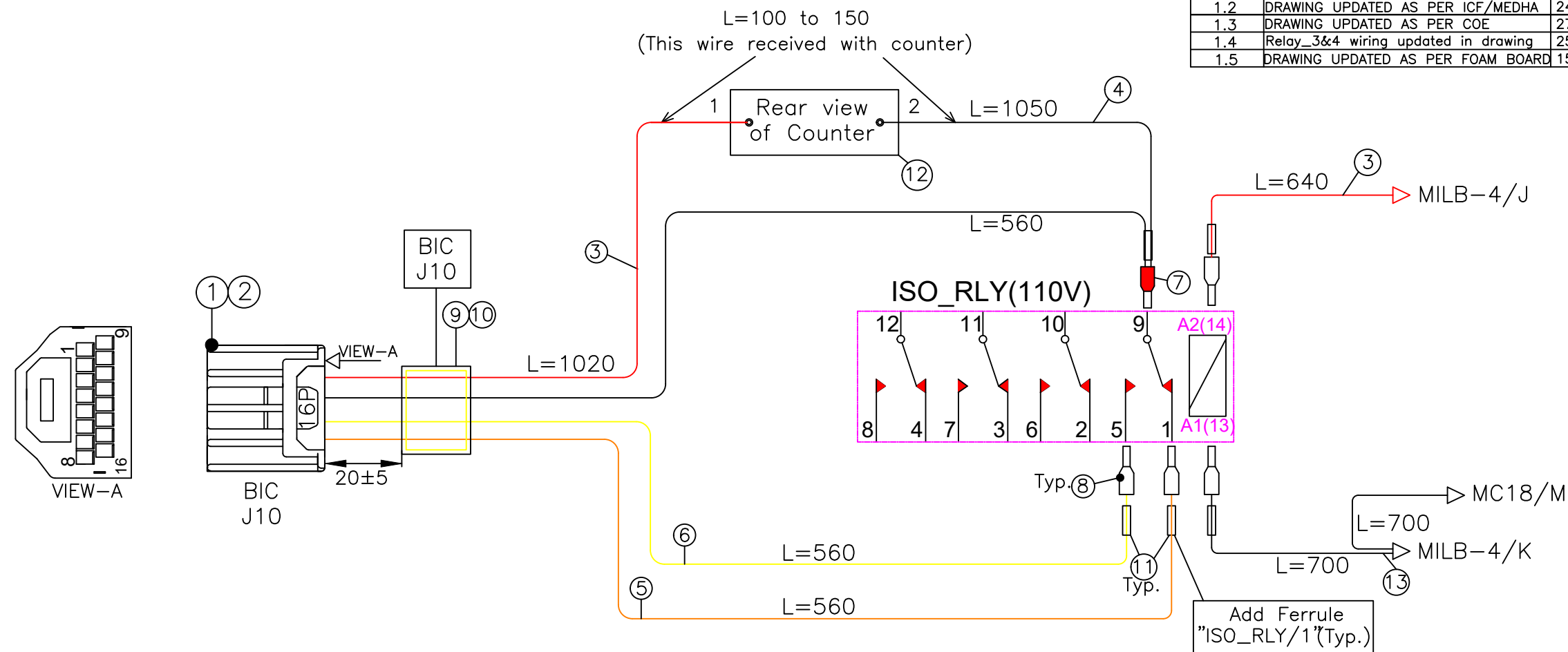
Note:—

1.Connect wires to respective relay as per the drawing

2.Add the labels to the wires at crimp terminals.

PROJECT: KAVACH		HBL HBL Engineering Ltd.		NAME		DESIGNATION		SIGN.	
ASSEMBLY: RIB-3Ph EMU				DESIGN		D.Lovaraju		Dy.Manager	
				CHECKED		J.V.Prasad		Sr.Manager	
				APPROVED		Y.Subramanyam		GM (Engg.)	
SCALE: NTS	SIZE: A3	TITLE: HARNS_INTRNAL_RIB_3PH EMU_MEMU HARNS ASY_RELAY LED INDICATIONS					DRAWING NO. 516490494		REV. 1.5
	SHEET: 10 OF 16						PART NO. 6000054517		
EFFECTIVE FROM: ALL DIMENSIONS ARE IN MM				TOLERANCE NOT SPECIFIED SHALL BE AS PER IS : 2102 – 1993. THIS DRAWING IS THE PROPERTY OF HBL ENGINEERING LIMITED AND MUST NOT BE COPIED, REPRODUCED, TRANSMITTED OR DISCLOSED TO ANY THIRD PARTY WITHOUT THE WRITTEN CONSENT OF THE COMPANY.					

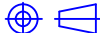
REV.NO.	CHANGE DETAILS	DATE	PREPARED	CHECKED	APPD.
1.1	UPDATED AS PER CoE/ICF REQUIREMENT	04.06.26	DLR	JVP	YS
1.2	DRAWING UPDATED AS PER ICF/MEDHA	24.06.26	DLR	JVP	YS
1.3	DRAWING UPDATED AS PER COE	27.07.26	DLR	JVP	YS
1.4	Relay_3&4 wiring updated in drawing	25.08.26	PGR	DLR	YS
1.5	DRAWING UPDATED AS PER FOAM BOARD	15.09.26	PGR	DLR	YS



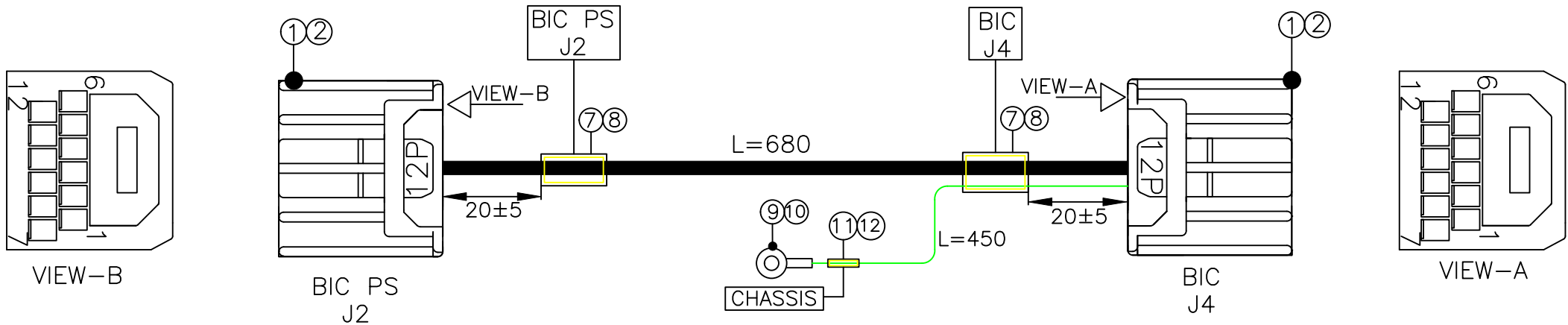
PART LIST				
ITEM NO/REF	QTY	UNITS	ITEM CODE	DESCRIPTION
1	1.00	PC	1000025256	CONNECTOR_SOCKET_16POS_PNO:MX34016SF1
2	4.00	PC	1000025249	CONNECTOR_CRIMP_TERMINAL_MX34
3	1.68	MTR	1000025384	WIRE_PTFE_RED_AWG 22_MIL-16878E/4BFB-2
4	3.10	MTR	1000013442	PTFE_WIRE_AWG 22(7/30)_BLACK
5	5.60	MTR	1000025383	WIRE_PTFE_ORANGE_AWG22_MIL-16878E/4BFB-3
6	5.60	MTR	1000025386	WIRE_PTFE_YELLOW_AWG 22_MIL16878E/4BFB-4
7	2.00	PC	1000095965	FERRULE_INSULATED_1MM_58-510L8_JAINSON
8	3.00	PC	1000095964	FERRULE_INSULATED_0.5MM_58-508L8_JAINSON
9	0.05	MTR	1000075057	HEAT_SHRINKABLE_SLEEVE_5MM_YELLOW
10	0.06	MTR	1000075058	HEAT_SHRINKABLE_SLEEVE_5MM_CLEAR
11	0.20	MTR	1000007218	FERRULE_TUBE_3.3MM_WHITE
12	1.00	PC	1000027376	EM_COUNTER_7 DIGIT_24V DC_KUBLER
13	0.03	MTR	1000016883	SLEEVE_HEAT SHINKABLE_DIA 3MM_BLACK

WIRE TERMINATION								
BIC J10	ISO RELAY	COUNTER	MILB-4	MC18	SIGNAL	WIRE COLOUR	BOM REF#	LENGTH
14	1	-	-	-	ISO_SW_N01	ORANGE	5	560
13	5	-	-	-	ISO_SW_NC1	YELLOW	6	560
11	-	1	-	-	COUNTER COIL	RED	3	1020
10	9	2	-	-	GND_24V	BLACK	4	560+1050
-	A2(14)	-	J	-	110V_DC(ISO)	RED	3	640
-	A1(13)	-	K	M	110V_GND	BLACK	4	700+700

Note:-
1. Label the all connectors as per respective location.
2. Counter wires will be soldered at the time of unit assembly.

PROJECT: KAVACH		HBL HBL Engineering Ltd.	NAME	DESIGNATION	SIGN.	
ASSEMBLY: RIB-3Ph EMU			DESIGN	D.Lovaraju	Dy.Manager	
			CHECKED	J.V.Prasad	Sr.Manager	
			APPROVED	Y.Subramanyam	GM (Engg.)	
SCALE: NTS	SIZE: A3	TITLE: HARNS_INTRNAL_RIB_3PH EMU_MEMU			DRAWING NO. 516490494	REV. 1.5
	SHEET: 11 OF 16	HARNS_ASSY_ISO SWITCH & COUNTER			PART NO. 6000054517	
EFFECTIVE FROM: ALL DIMENSIONS ARE IN MM				TOLERANCE NOT SPECIFIED SHALL BE AS PER IS : 2102 - 1993. THIS DRAWING IS THE PROPERTY OF HBL ENGINEERING LIMITED AND MUST NOT BE COPIED, REPRODUCED, TRANSMITTED OR DISCLOSED TO ANY THIRD PARTY WITHOUT THE WRITTEN CONSENT OF THE COMPANY.		

REV.NO.	CHANGE DETAILS	DATE	PREPARED	CHECKED	APPD.
1.1	UPDATED AS PER CoE/ICF REQUIREMENT	04.06.26	DLR	JVP	YS
1.2	DRAWING UPDATED AS PER ICF/MEDHA	24.06.26	DLR	JVP	YS
1.3	DRAWING UPDATED AS PER COE	27.07.26	DLR	JVP	YS
1.4	Relay_3&4 wiring updated in drawing	25.08.26	PGR	DLR	YS
1.5	DRAWING UPDATED AS PER FOAM BOARD	15.09.26	PGR	DLR	YS



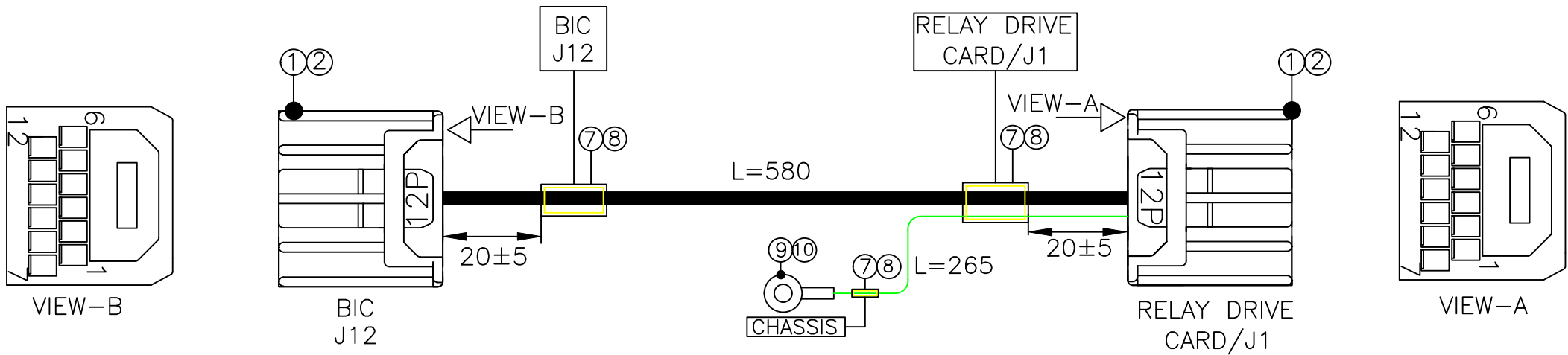
PART LIST					
ITEM NO/REF	QTY	UNITS	ITEM CODE	DESCRIPTION	
1	2.00	PC	1000027692	CONNECTOR SOCKET 12POS MX34012SF1	
2	21.00	PC	1000025249	CONNECTOR_CRIMP_TERMINAL_MX34	
3	2.72	MTR	1000025384	WIRE_PTFE_RED_AWG 22_MIL-16878E/4BFB-2	
4	2.72	MTR	1000013442	PTFE_WIRE_AWG 22(7/30)_BLACK	
5	0.68	MTR	1000025379	WIRE_PTFE_BLUE_AWG 22_MIL-16878E/4BFB-6	
6	1.13	MTR	1000025381	WIRE_PTFE_GREEN_AWG 22_MIL16878E/4BFB-5	
7	0.10	MTR	1000016947	SLEEVE_MARKER_3/8"_YELLOW	
8	0.15	MTR	1000016866	SLEEVE_CLEAR_3/8"	
9	1.00	PC	1000010068	LUG_1.5SQ MM_CU_RING TYPE M6	
10	0.10	MTR	1000016883	SLEEVE_HEAT SHINKABLE_DIA 3MM_BLACK	
11	0.05	MTR	1000075057	HEAT_SHRINKABLE_SLEEVE_5MM_YELLOW	
12	0.06	MTR	1000075058	HEAT_SHRINKABLE_SLEEVE_5MM_CLEAR	

WIRE TERMINATION					
BIC PS J2	BIC J4	SIGNAL	WIRE COLOUR	BOM REF#	LENGTH
1	1	24V_ISO	RED	3	680
2	2	24V_ISO	RED	3	
7	7	24V_ISO	RED	3	
8	8	24V_ISO	RED	3	
3	3	GND_ISO	BLACK	4	
4	4	GND_ISO	BLACK	4	
5	5	GND_ISO	BLACK	4	
10	10	GND_ISO	BLACK	4	
6	6	A1	BLUE	5	
12	12	K1	GREEN	6	
-	9	CHASSIS	GREEN	6	450

Note:-
1. Label the all connectors as per respective location.

PROJECT: KAVACH		HBL HBL Engineering Ltd.		NAME		DESIGNATION		SIGN.		
ASSEMBLY: RIB-3Ph EMU				DESIGN		D.Lovaraju		Dy.Manager		
				CHECKED		J.V.Prasad		Sr.Manager		
				APPROVED		Y.Subramanyam		GM (Engg.)		
SCALE: NTS	SIZE: A3	TITLE: HARNS_INTRNAL_RIB_3PH EMU_MEMU HARNS ASY_24-ISO POWER SUPPLY					DRAWING NO. 516490494		REV. 1.5	
	SHEET: 12 OF 16						PART NO. 6000054517			
EFFECTIVE FROM: ALL DIMENSIONS ARE IN MM				TOLERANCE NOT SPECIFIED SHALL BE AS PER IS : 2102 - 1993. THIS DRAWING IS THE PROPERTY OF HBL ENGINEERING LIMITED AND MUST NOT BE COPIED, REPRODUCED, TRANSMITTED OR DISCLOSED TO ANY THIRD PARTY WITHOUT THE WRITTEN CONSENT OF THE COMPANY.						

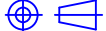
REV.NO.	CHANGE DETAILS	DATE	PREPARED	CHECKED	APPD.
1.1	UPDATED AS PER CoE/ICF REQUIREMENT	04.06.26	DLR	JVP	YS
1.2	DRAWING UPDATED AS PER ICF/MEDHA	24.06.26	DLR	JVP	YS
1.3	DRAWING UPDATED AS PER COE	27.07.26	DLR	JVP	YS
1.4	Relay_3&4 wiring updated in drawing	25.08.26	PGR	DLR	YS
1.5	DRAWING UPDATED AS PER FOAM BOARD	15.09.26	PGR	DLR	YS



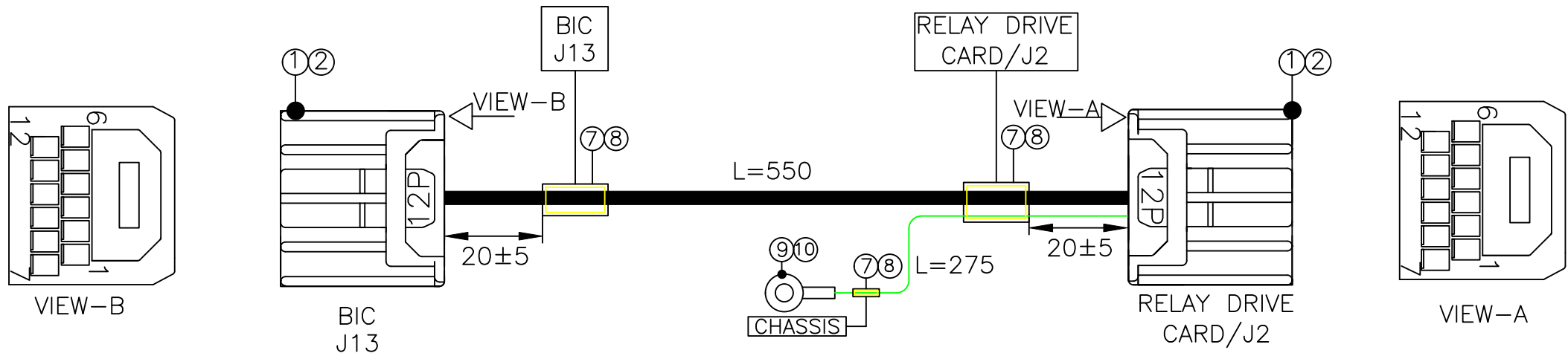
PART LIST				
ITEM NO/REF	QTY	UNITS	ITEM CODE	DESCRIPTION
1	2.00	PC	1000027692	CONNECTOR SOCKET 12POS MX34012SF1
2	12.00	PC	1000025249	CONNECTOR_CRIMP_TERMINAL_MX34
3	1.16	MTR	1000025384	WIRE_PTFE_RED_AWG 22_MIL-16878E/4BFB-2
4	1.16	MTR	1000013442	PTFE_WIRE_AWG 22(7/30)_BLACK
5	0.58	MTR	1000025379	WIRE_PTFE_BLUE_AWG 22_MIL-16878E/4BFB-6
6	0.28	MTR	1000025381	WIRE_PTFE_GREEN_AWG 22_MIL16878E/4BFB-5
7	0.10	MTR	1000075057	HEAT_SHRINKABLE_SLEEVE_5MM_YELLOW
8	0.15	MTR	1000075058	HEAT_SHRINKABLE_SLEEVE_5MM_CLEAR
9	1.00	PC	1000010068	LUG_1.5SQ MM_CU_RING TYPE M6
10	0.05	MTR	1000016883	SLEEVE_HEAT SHINKABLE_DIA 3MM_BLACK

WIRE TERMINATION					
BIC J12	RELAY DRIVE CARD/J1	SIGNAL	WIRE COLOUR	BOM REF#	LENGTH
1	1	24V_ISO	RED	3	580
2	2	GND_ISO	BLACK	4	
7	7	24V_ISO	RED	3	
8	8	GND_ISO	BLACK	4	
4	4	FOULING_MARK	BLUE	5	265
-	6	CHASSIS	GREEN	6	

Note:-
1. Label the all connectors as per respective location.

PROJECT: KAVACH		HBL HBL Engineering Ltd.		NAME	DESIGNATION	SIGN.
ASSEMBLY: RIB-3Ph EMU			DESIGN	D.Lovaraju	Dy.Manager	
			CHECKED	J.V.Prasad	Sr.Manager	
			APPROVED	Y.Subramanyam	GM (Engg.)	
SCALE: NTS	SIZE: A3	TITLE: HARNS_INTRNAL_RIB_3PH EMU_MEMU HARNS ASY_FOULING MARK INPUT			DRAWING NO. 516490494	REV. 1.5
	SHEET: 13 OF 16				PART NO. 6000054517	
EFFECTIVE FROM: ALL DIMENSIONS ARE IN MM			TOLERANCE NOT SPECIFIED SHALL BE AS PER IS : 2102 - 1993. THIS DRAWING IS THE PROPERTY OF HBL ENGINEERING LIMITED AND MUST NOT BE COPIED, REPRODUCED, TRANSMITTED OR DISCLOSED TO ANY THIRD PARTY WITHOUT THE WRITTEN CONSENT OF THE COMPANY.			

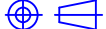
REV.NO.	CHANGE DETAILS	DATE	PREPARED	CHECKED	APPD.
1.1	UPDATED AS PER CoE/ICF REQUIREMENT	04.06.26	DLR	JVP	YS
1.2	DRAWING UPDATED AS PER ICF/MEDHA	24.06.26	DLR	JVP	YS
1.3	DRAWING UPDATED AS PER COE	27.07.26	DLR	JVP	YS
1.4	Relay_3&4 wiring updated in drawing	25.08.26	PGR	DLR	YS
1.5	DRAWING UPDATED AS PER FOAM BOARD	15.09.26	PGR	DLR	YS



PART LIST				
ITEM NO/REF	QTY	UNITS	ITEM CODE	DESCRIPTION
1	2.00	PC	1000027692	CONNECTOR SOCKET 12POS MX34012SF1
2	12.00	PC	1000025249	CONNECTOR_CRIMP_TERMINAL_MX34
3	1.10	MTR	1000025384	WIRE_PTFE_RED_AWG 22_MIL-16878E/4BFB-2
4	1.10	MTR	1000013442	PTFE_WIRE_AWG 22(7/30)_BLACK
5	0.55	MTR	1000025379	WIRE_PTFE_BLUE_AWG 22_MIL-16878E/4BFB-6
6	0.29	MTR	1000025381	WIRE_PTFE_GREEN_AWG 22_MIL16878E/4BFB-5
7	0.10	MTR	1000075057	HEAT_SHRINKABLE_SLEEVE_5MM_YELLOW
8	0.15	MTR	1000075058	HEAT_SHRINKABLE_SLEEVE_5MM_CLEAR
9	1.00	PC	1000010068	LUG_1.5SQ MM_CU_RING TYPE M6
10	0.05	MTR	1000016883	SLEEVE_HEAT SHINKABLE_DIA 3MM_BLACK

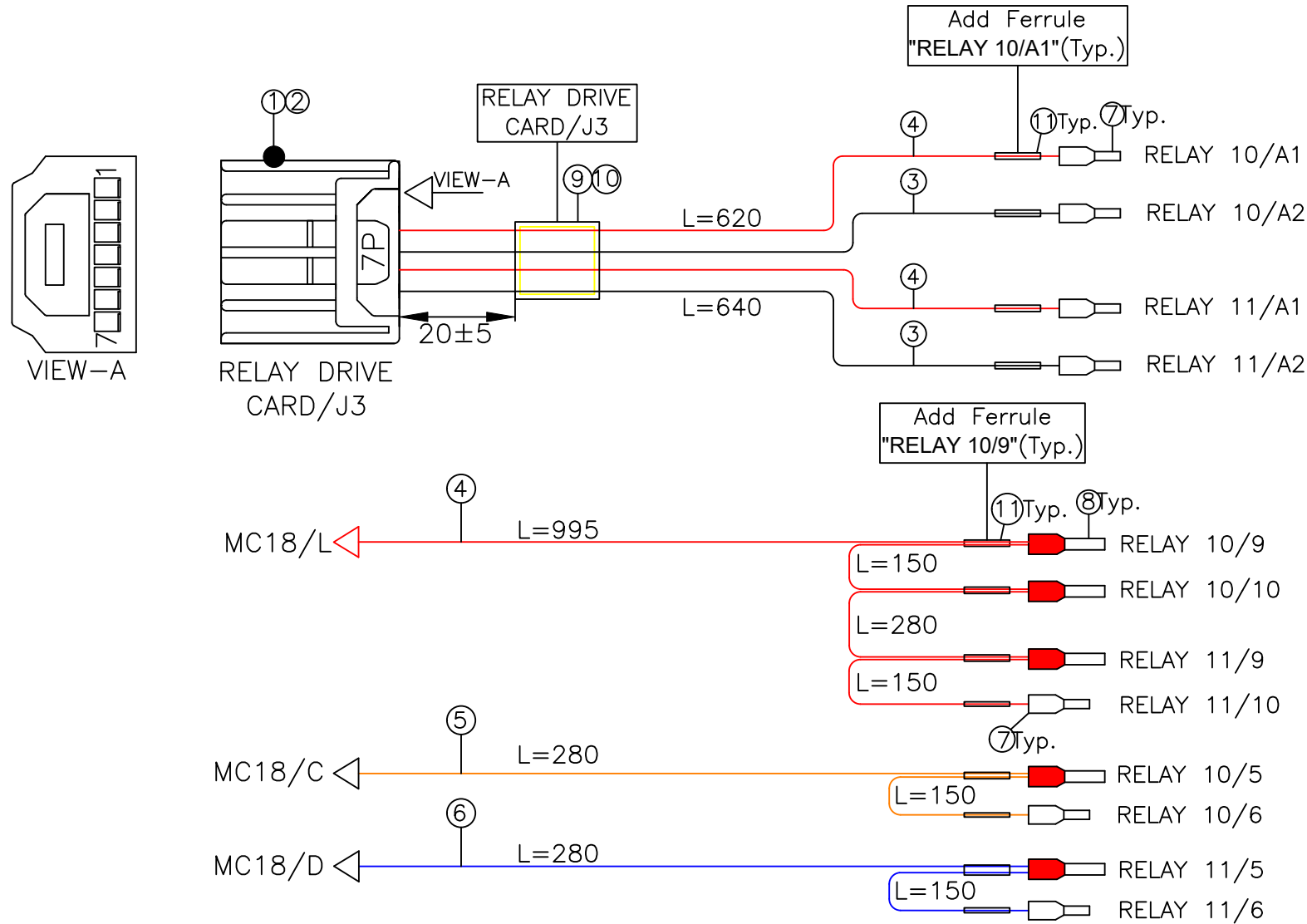
WIRE TERMINATION					
BIC J13	RELAY DRIVE CARD/J2	SIGNAL	WIRE COLOUR	BOM REF#	LENGTH
1	1	24V_ISO	RED	3	550
2	2	GND_ISO	BLACK	4	
7	7	24V_ISO	RED	3	
8	8	GND_ISO	BLACK	4	
4	4	NEUTRAL_SEC	BLUE	5	275
-	6	CHASSIS	GREEN	6	

Note:-
1. Label the all connectors as per respective location.

PROJECT: KAVACH		HBL HBL Engineering Ltd.		NAME	DESIGNATION	SIGN.
ASSEMBLY: RIB-3Ph EMU			DESIGN	D.Lovaraju	Dy.Manager	
			CHECKED	J.V.Prasad	Sr.Manager	
			APPROVED	Y.Subramanyam	GM (Engg.)	
SCALE: NTS	SIZE: A3	TITLE: HARNS_INTRNAL_RIB_3PH EMU_MEMU HARNS ASY_NEUTRAL SECTION INPUT			DRAWING NO. 516490494	REV. 1.5
	SHEET: 14 OF 16				PART NO. 6000054517	
EFFECTIVE FROM: ALL DIMENSIONS ARE IN MM			TOLERANCE NOT SPECIFIED SHALL BE AS PER IS : 2102 - 1993. THIS DRAWING IS THE PROPERTY OF HBL ENGINEERING LIMITED AND MUST NOT BE COPIED, REPRODUCED, TRANSMITTED OR DISCLOSED TO ANY THIRD PARTY WITHOUT THE WRITTEN CONSENT OF THE COMPANY.			

REV.NO.	CHANGE DETAILS	DATE	PREPARED	CHECKED	APPD.
1.1	UPDATED AS PER CoE/ICF REQUIREMENT	04.06.26	DLR	JVP	YS
1.2	DRAWING UPDATED AS PER ICF/MEDHA	24.06.26	DLR	JVP	YS
1.3	DRAWING UPDATED AS PER COE	27.07.26	DLR	JVP	YS
1.4	Relay_3&4 wiring updated in drawing	25.08.26	PGR	DLR	YS
1.5	DRAWING UPDATED AS PER FOAM BOARD	15.09.26	PGR	DLR	YS

P. Ganesan
D. L. Raju

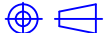


WIRE TERMINATION-1						
RELAY DRIVE CARD/J3	RELAY10	RELAY11	SIGNAL	WIRE COLOUR	BOM REF#	LENGTH
1	A1	—	24V_DC	RED	4	620
3	A2	—	FLNG_MRK_COIL	BLACK	3	
5	—	A1	24V_DC	RED	4	640
7		A2	NTRL_SEC_COIL	BLACK	3	

WIRE TERMINATION-2						
MC18	RELAY10	RELAY11	SIGNAL	WIRE COLOUR	BOM REF#	LENGTH
C	5&6	—	FOULING_MARK	ORANGE	5	280+150
D	—	5&6	NUETRAL_SECTION	BLUE	6	280+150
L	9&10	9&10	LOCO_+VE	RED	4	1575

PART LIST				
ITEM NO/REF	QTY	UNITS	ITEM CODE	DESCRIPTION
1	1.00	PC	1000041208	CONNECTOR_SOCKET_7POS_MX34007SF1
2	4.00	PC	1000025249	CONNECTOR_CRIMP_TERMINAL_MX34
3	1.37	MTR	1000013442	PTFE_WIRE_AWG 22(7/30)_BLACK
4	2.84	MTR	1000025384	WIRE_PTFE_RED_AWG 22_MIL-16878E/4BFB-2
5	0.45	MTR	1000025383	WIRE_PTFE_ORANGE_AWG22_MIL-16878E/4BFB-3
6	0.45	MTR	1000025379	WIRE_PTFE_BLUE_AWG 22_MIL-16878E/4BFB-6
7	7.00	PC	1000095964	FERRULE_INSULATED_0.5MM_58-508L8_JAINSON
8	5.00	PC	1000095965	FERRULE_INSULATED_1MM_58-510L8_JAINSON
9	0.05	MTR	1000075057	HEAT_SHRINKABLE_SLEEVE_5MM_YELLOW
10	0.06	MTR	1000075058	HEAT_SHRINKABLE_SLEEVE_5MM_CLEAR
11	0.20	MTR	1000007218	FERRULE_TUBE_3.3MM_WHITE

Note:—
1. Label the all connectors as per respective location.

PROJECT: KAVACH		HBL HBL Engineering Ltd.		NAME	DESIGNATION	SIGN.
ASSEMBLY: RIB-3Ph EMU			DESIGN	D.Lovaraju	Dy.Manager	
			CHECKED	J.V.Prasad	Sr.Manager	
			APPROVED	Y.Subramanyam	GM (Engg.)	
SCALE: NTS	SIZE: A3	TITLE: HARNS_INTRNAL_RIB_3PH EMU_MEMU HARNS ASY_FLNG MARK&NTRL SECTION			DRAWING NO. 516490494	REV. 1.5
	SHEET: 15 OF 16				PART NO. 6000054517	
EFFECTIVE FROM: ALL DIMENSIONS ARE IN MM			TOLERANCE NOT SPECIFIED SHALL BE AS PER IS : 2102 - 1993. THIS DRAWING IS THE PROPERTY OF HBL ENGINEERING LIMITED AND MUST NOT BE COPIED, REPRODUCED, TRANSMITTED OR DISCLOSED TO ANY THIRD PARTY WITHOUT THE WRITTEN CONSENT OF THE COMPANY.			

5

4

3

2

1

REV.NO.	CHANGE DETAILS	DATE	PREPARED	CHECKED	APPD.
1.1	UPDATED AS PER CoE/ICF REQUIREMENT	04.06.26	DLR	JVP	YS
1.2	DRAWING UPDATED AS PER ICF/MEDHA	24.06.26	DLR	JVP	YS
1.3	DRAWING UPDATED AS PER COE	27.07.26	DLR	JVP	YS
1.4	Relay_3&4 wiring updated in drawing	25.08.26	PGR	DLR	YS
1.5	DRAWING UPDATED AS PER FOAM BOARD	15.09.26	PGR	DLR	YS

P. Ganesan

D. L. Raju

5

4

3

2

1

ITEM NO/REF

QTY

UNITS

ITEM CODE

DESCRIPTION

MANUFACTURER

PART NO.

1

2.00

PC

1000025257

CONNECTOR_SOCKET_20POS_PNO:MX34020SF1

JAE CONNECTORS

MX34020SF1

2

2.00

PC

1000025256

CONNECTOR_SOCKET_16POS_PNO:MX34016SF1

JAE CONNECTORS

MX34016SF1

3

8.00

PC

1000027692

CONNECTOR_SOCKET_12POS_MX34012SF1

JAE CONNECTORS

MX34012SF1

4

3.00

PC

1000041208

CONNECTOR_SOCKET_7POS_MX34007SF1

JAE CONNECTORS

MX34007SF1

5

2.00

PC

1000027695

CONNECTOR_SOCKET_5POS_MX34005SF1

JAE CONNECTORS

MX34005SF1

6

119.0

PC

1000025249

CONNECTOR_CRIMP_TERMINAL_MX34

JAE CONNECTORS

M34S75C4F2

7

1.00

PC

1000050114

CONNECTOR_PLUG_19PIN_62IN-5024-14-19P

AMPHENOL

62IN50241214P

8

1.00

PC

1000088342

CONNECTOR_RECEP_14P_62IN-5024-14-A14-P

AMPHENOL

62IN502414A14P

9

1.00

PC

1000088343

CONNECTOR_RECEP_14P_62IN-5024-14-A14-S

AMPHENOL

62IN502414A14S

10

1.00

PC

1000088346

CONNECTOR_RECEP_12P_62IN-5024-14-12P

AMPHENOL

62IN50241412P

11

2.00

PC

1000088347

CONNECTOR_RECEP_12P_62IN-5024-14-12S

AMPHENOL

62IN50241412S

12

3.35

MTR

1000016883

SLEEVE_HEAT SHINKABLE_DIA 3MM_BLACK

TYCO

-

13

25.87

MTR

1000013442

PTFE_WIRE_AWG 22(7/30)_BLACK

Tonk & Associates

-

14

3.00

MTR

1000025380

WIRE_PTFE_BROWN_AWG 22_MIL-16878E/4BFB-1

Tonk & Associates

-

15

25.50

MTR

1000025384

WIRE_PTFE_RED_AWG 22_MIL-16878E/4BFB-2

Tonk & Associates

-

16

7.00

MTR

1000025383

WIRE_PTFE_ORANGE_AWG22_MIL-16878E/4BFB-3

Tonk & Associates

-

17

4.30

MTR

1000025386

WIRE_PTFE_YELLOW_AWG 22_MIL16878E/4BFB-4

Tonk & Associates

-

18

5.20

MTR

1000025381

WIRE_PTFE_GREEN_AWG 22_MIL16878E/4BFB-5

Tonk & Associates

-

19

2.60

MTR

1000025379

WIRE_PTFE_BLUE_AWG 22_MIL-16878E/4BFB-6

Tonk & Associates

-

20

3.50

MTR

1000025385

WIRE_PTFE_VIOLET_AWG22_MIL-16878E/4BFB-7

Tonk & Associates

-

21

3.30

MTR

1000025382

WIRE_PTFE_GREY_AWG 22_MIL-16878E/4BFB-8

Tonk & Associates

-

22

1.50

MTR

1000026453

PTFE_WIRE_AWG 22(7/30)_WHITE

Tonk & Associates

-

23

0.10

MTR

1000017038

SLEEVE_SHRINKABLE_1/16_MX446320_BLACK

-

-

24

5.00

PC

1000010068

LUG_1.5SQ MM_CU_RING TYPE M6

Reputed

-

25

26.00

PC

1000095965

FERRULE_INSULATED_1MM_58-510L8_JAINSON

Reputed

-

26

31.00

PC

1000095964

FERRULE_INSULATED_0.5MM_58-508L8_JAINSON

Reputed

-

27

5.00

PC

1000075399

DIODE_SCHOTTKY_LEADED_40V_5A

Reputed

-

28

0.20

MTR

1000026054

SLEEVE_HEAT SHRINKABLE_5MM_BLACK

Reputed

-

29

1.65

MTR

1000075057

HEAT_SHRINKABLE_SLEEVE_5MM_YELLOW

WOER

-

30

1.86

MTR

1000075058

HEAT_SHRINKABLE_SLEEVE_5MM_CLEAR

WOER

-

31

0.35

MTR

1000074458

HEAT-SHRINKABLE_SLEEVE_16MM_BLACK

WOER

-

32

2.30

MTR

1000007218

FERRULE_TUBE_3.3MM_WHITE

WOER

-

33

2.05

MTR

1000016947

SLEEVE_MARKER_3/8" _YELLOW

WOER

-

34

0.86

MTR

1000016866

SLEEVE_CLEAR_3/8"

WOER

-

35

1.00

PC

1000027376

EM_COUNTER_7 DIGIT_24V DC_KUBLER

-

-

36

6.00

PC

1000037004

FERRULE_INSULATED_1.5mm_58-512L8_JAINSON

-

-

37

0.10

MTR

1000017043

SLEEVE_SHRINKABLE_3/32_BLACK

-

-

38

100.00

PC

1000001886

CABLE TIES FR TYPE_100MM

-

-

39

4.00

PC

1000006449

DIODE_LEADED_1N4007_DO-41

PROJECT:
KAVACH

ASSEMBLY:
RIB-3Ph EMU

SCALE:
NTS

SIZE:
A3

SHEET:
16 OF 16

EFFECTIVE FROM:
ALL DIMENSIONS ARE IN MM

HBL

HBL Engineering Ltd.

TITLE:
HARNS_INTRNAL_RIB_3PH EMU_MEMU

BOM SUMMARY SHEET

NAME
D.Lovaraju

DESIGNATION
Dy.Manager

NAME
J.V.Prasad

DESIGNATION
Sr.Manager

NAME
Y.Subramanyam

DESIGNATION
GM (Engg.)

DRAWING NO.
516490494

PART NO.
6000054517

REV.
1.5

TOLERANCE NOT SPECIFIED SHALL BE AS PER IS : 2102 - 1993.

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5

4

3

2

1

EG-EN-FT-07 REV:1

Signature: P Ganeswararao
P Ganeswararao (Sep 19, 2026 15:57:17 GMT+5.5)
Email: ganeswararao.palli@hbl.in
Title: Engineer

Signature: 
Subrahmanyam Y (Sep 19, 2026 16:29:41 GMT+5.5)
Email: subrahmanyam.yadavalli@hbl.in
Title: GM - Engineering

Signature: D L Raju
D L Raju (Sep 19, 2026 16:08:38 GMT+5.5)
Email: lovaraju.dadala@hbl.in
Title: Dy Manager

6000054517_HARNS_INTRNAL_RIB_3PH EMU_MEMU_516490494

Final Audit Report

2026-09-19

Created:	2026-09-19 (India Standard Time)
By:	JAGATHA PRASAD (prasad.jagatha@hbl.in)
Status:	Signed
Transaction ID:	CBJCHBCAABAAM_RiMIsE8OeCYBTp00pidBCVFRqdkt4

"6000054517_HARNS_INTRNAL_RIB_3PH EMU_MEMU_516490494" History

-  Document created by JAGATHA PRASAD (prasad.jagatha@hbl.in)
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-  Document emailed to ganeswararao.palli@hbl.in for signature
2026-09-19 - 3:53:51 PM GMT+5.5
-  Email viewed by ganeswararao.palli@hbl.in
2026-09-19 - 3:54:45 PM GMT+5.5- IP address: 122.187.81.114
-  Signer ganeswararao.palli@hbl.in entered name at signing as P Ganeswararao
2026-09-19 - 3:57:15 PM GMT+5.5- IP address: 122.187.81.114
-  Document e-signed by P Ganeswararao (ganeswararao.palli@hbl.in)
Signature Date: 2026-09-19 - 3:57:17 PM GMT+5.5 - Time Source: server- IP address: 122.187.81.114 - Signature Appearance Selected: TYPE
-  Document emailed to lovaraju.dadala@hbl.in for signature
2026-09-19 - 3:57:23 PM GMT+5.5
-  Email viewed by lovaraju.dadala@hbl.in
2026-09-19 - 4:07:26 PM GMT+5.5- IP address: 106.77.160.61
-  Signer lovaraju.dadala@hbl.in entered name at signing as D L Raju
2026-09-19 - 4:08:36 PM GMT+5.5- IP address: 106.77.160.61
-  Document e-signed by D L Raju (lovaraju.dadala@hbl.in)
Signature Date: 2026-09-19 - 4:08:38 PM GMT+5.5 - Time Source: server- IP address: 106.77.160.61 - Signature Appearance Selected: MOBILE_TYPE



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Agreement completed.

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